

Risk communication and community engagement readiness and response toolkit

Substandard and falsified medicines



World Health
Organization

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All external experts who reviewed the toolkit completed a WHO declaration of interest to disclose potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence concerning the subject. WHO reviewed these and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects reviewed.

Abbreviations and acronyms

CSO	civil society organization
GSMS	Global Surveillance and Monitoring System (WHO)
IFRC	International Federation of Red Cross and Red Crescent Societies
IHR	International Health Regulations
IMST	incident management support team
LMICs	low- and middle-income countries
MEL	measurement, evaluation and learning
NMRA	national medicines regulatory authority
NRA	national regulatory authority
NRRAs	national and regional regulatory authorities
PESTEL	political, economic, sociological, technological, legal and environmental
RCCE	risk communication and community engagement
SOCO	single overarching communications outcome
UN	United Nations
UNICEF	United Nations Children's Fund
UNODC	UN Office on Drugs and Crime
WHO	World Health Organization

Glossary

Adverse event	Any unexpected or harmful medical occurrence that happens to a patient during medical treatment or a clinical trial, related to care provided. Unlike direct side-effects, an adverse event does not necessarily mean the medication directly caused the problem.
Authorized medicines	Also known as licensed or approved drugs, these are medicinal products that have been granted marketing approval by a regulatory authority after a thorough assessment of their safety, efficacy and quality.
Behavioural insights	Information about variables that influence behaviours at the individual, community and population level and can improve the design of policies and programmes, communications, and products and services to achieve better health for all.
Behavioural science	The multidisciplinary study of human behaviour, encompassing how individuals think, feel and act in various contexts. It draws on fields like psychology, sociology and economics to understand, predict and influence behaviour and determinants of behaviour, often using experimental methods and observational data. It is applied in protecting and improving people's health by informing the development of public health policies, programmes and interventions.
Collective service for risk communication and community engagement (RCCE)	A partnership between the International Federation of Red Cross and Red Crescent Societies (IFRC), United Nations Children's Fund (UNICEF), WHO and the Global Outbreak Alert and Response Network (GOARN), as well as key stakeholders from the public health and humanitarian sectors.
Community	Refers to a group of people connected by common characteristics, such as geographic location, age, gender, profession, ethnicity, faith, shared vulnerability or risk, or shared interests and values.
Community engagement	The collaborative process that involves people in understanding the risks they face and includes communities in developing health and response practices that are acceptable and workable for them. The goal of community engagement is to empower communities and to develop shared leadership throughout the emergency response cycle.
Emergency	An emergency is an urgent, unexpected and usually dangerous situation that poses an immediate risk to health, life, property or environment and requires immediate action.
Falsified medical product	A medical product that deliberately and fraudulently misrepresents its identity, composition or source.
Global Surveillance and Monitoring System (GSMS)	A system through which Member States and others submit reports of suspected or validated substandard and falsified medical products. The GSMS facilitates high-quality reporting of substandard and falsified medical products and ensures that the collected data are analysed and used to inform policies, procedures and processes aimed at protecting public health at national, regional and global levels.
Health emergency management cycle	Spans the prevention, preparedness, readiness, response and recovery phases of health emergencies that all organizations and governments should follow to reduce the impact of disease outbreaks, health emergencies and disasters. Countries and communities may be engaged in different phases for multiple outbreaks and emergencies simultaneously.

Infodemic	An infodemic is an overabundance of information, accurate or not, in the digital and physical environment, accompanying an acute health event such as a disease outbreak or epidemic.
Informal market of medical products	A sector of the national or local economy where the manufacture, import or export, distribution, sale, supply or purchase of medical products takes place outside of the legal, regulatory or administrative oversight of relevant public health or regulatory authorities. The medical products available on this market may not have been assessed for safety, efficacy or quality by public health and regulatory authorities or the assessment may not have been conducted by appropriately qualified persons or entities.
National regulatory system	The national regulatory authority (NRA) is the institution responsible for assuring the quality, safety and efficacy of health products, as well as the relevance and accuracy of product information. The number of regulatory entities responsible for different regulatory functions may vary from one country to another.
Outbreak	A sudden increase in occurrences of a disease when cases are in excess of normal expectations for the location or season.
Readiness	Refers to the ability of countries, communities and organizations to be able to respond quickly and effectively to health emergencies from any hazard. Operational readiness is a critical enabler of resilience in communities and health systems, helping them to withstand crisis. Fast-tracking, activating, testing or prepositioning specific functional capabilities are all important functions for enhanced readiness.
Regulatory authority	A government authority or other entity with the legal mandate to regulate the use and sale of medical products within its jurisdiction. It may also take enforcement actions to ensure that all marketed medical products comply with applicable legal requirements.
Response	The phase of a health emergency or outbreak activated once the hazard, risk or threat hits, with the implementation of life-saving public health and health interventions to save lives and protect the most vulnerable.
Risk assessment	Risk assessment is a systematic and continuous process of gathering, evaluating and documenting information to manage and mitigate the impact of acute public health threats. It consists of the identification of hazards and the evaluation of the risk associated with exposure to those hazards.
Risk communication	Real-time exchange of information, advice and opinions between experts and people who are facing a risk or threat to their health, social or economic well-being. The purpose of risk communication is to provide people with accurate and timely information and to support them in making informed decisions to mitigate the effects of a threat or hazard.
Stakeholders	Any individual, group or organization that can affect, be affected by, or perceive itself to be affected by a risk (posed by substandard and falsified medicines). Primary stakeholders include and are not limited to patients, health care professionals, medicines regulatory authorities and the pharmaceutical industry, and governments and community leaders, which have a vested interest in protecting the health of their own country, region or community.
Substandard medicines	Authorized medicines that fail to meet quality standards or specifications. They are often due to poor manufacturing practices, inadequate quality control or improper storage and handling.
Unregistered/unlicensed medicines	Medicines that have not undergone evaluation and/or approval by the national or regional regulatory authority for the market in which they are marketed, distributed or used, subject to conditions under national or regional regulation and legislation.

1. Overview of the risk communication and community engagement readiness and response toolkit

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This toolkit is a comprehensive set of practical tools and resources designed to support country-level risk communication and community engagement (RCCE) practitioners, national regulatory authorities (NRAs), decision-makers and partners in the planning and implementation of readiness and response activities for emergencies involving substandard and falsified medicines.

The toolkit contains:

- information about incidents involving substandard and falsified medicines;
- RCCE considerations for how to approach key issues during such incidents;
- tools for assessing risk and risk perception, as well as understanding the target audience(s);
- tools for collecting behavioural and social data to inform RCCE strategy development;
- methods to analyse data and turn findings into actionable insights for planning and implementing activities;
- case studies; and
- links to existing RCCE and resources, tools and trainings.

It is one of a suite of toolkits on RCCE readiness and response to a range of disease and public health emergencies.

The toolkit has been developed through an iterative, consultative process that has followed several steps to identify, collate and refine the information, tools and best practices it contains.

These steps included the following stages.

Literature review

An extensive review was conducted of the scientific and grey literature, research papers and published documents related to substandard, falsified and contaminated medicines, RCCE and health emergencies.

A structured search of online databases (PubMed, IRIS – Institutional Repository for Information Sharing, ReliefWeb and Google Scholar) was conducted to identify publications related to substandard and falsified medicines, and the role of RCCE during incidents involving them.

Keywords supplied by the technical teams were used as the foundation of the search to identify relevant documents, from which other specific terms and keywords were extracted. Documents were systematically reviewed for content on key thematic areas, methods and definitions relevant to the development of RCCE plans and strategies. This included but was not limited to behavioural science methods related to public health; stakeholder engagement; risk and situational analysis; audience analysis; risk assessment; community listening; risk communication strategies; best practices for message development; and measurement, evaluation and learning (MEL) frameworks and methodologies. Retrieved publications were assessed for relevance, uploaded to a database and logged into a tracking sheet, highlighting them for further consideration.

Iterative consultation

Following development of the toolkit and integration of relevant publications and sources in close consultation with WHO technical teams, the toolkit was then reviewed and revised by RCCE and substandard and falsified subject matter experts at country, regional and global levels. This was done through an iterative consultation process between September 2024 and October 2025.

Pilot testing

Tools within this toolkit underwent rigorous testing both in emergency response settings and during capacity-building activities.

Readiness and response within the context of substandard and falsified medicines

In recent years, WHO, Member States and partners have engaged in significant efforts to strengthen the architecture for health emergency prevention, preparedness, readiness, response and recovery. Readiness and response are closely connected. Readiness builds on the preparedness phase and is the interface between preparedness and immediate response to an emergency. WHO works constantly with regulators, manufacturers and other partners, including the public, to prevent, detect and respond to incidents related to substandard and falsified medicines. To support readiness in the context of substandard and falsified medicines and medical products, WHO has issued guidance and assessment tools on good governance and transparency, particularly in relation to the procurement of medicines and medical products (1).

National medicines regulatory authorities (NMRAs) are responsible for ensuring that products released for public distribution (normally pharmaceuticals and biological products, such as vaccines and medical devices, including test kits) are evaluated properly and meet international standards of quality, safety and efficacy (2).

WHO works with NMRAs in countries to implement prevention, detection and response strategies that promote securing the supply chains of the raw materials used to manufacture these products, bolster regulation, surveillance and detection, and raise awareness among stakeholders and the public of the existence and risks of substandard and falsified medicines. NMRAs are also key stakeholders in the response to incidents involving substandard and falsified medicines.

Experience has shown that countries that systematically ready their regulatory, health and emergency systems can respond more quickly, cohesively and equitably to a threat or

emergency, such as those involving contaminated medicines. With the addition of robust regulatory frameworks, mature national and regional capacities to ensure supply chain integrity and early warning, monitoring and surveillance, countries can curb the impact of these incidents and ultimately save lives. Strong collaboration and communication pathways between stakeholders, including national regulatory authorities, civil society, manufacturers, health care practitioners, laboratories, law enforcement and others are also essential. Local systems may include pharmacovigilance networks, customs and law enforcement collaboration, and public reporting platforms that allow health care professionals and citizens to report suspicious products.

An effective response requires early intervention and timely, proportionate and effective regulatory action, most often involving quarantine or seizure of the product, sampling and laboratory testing to confirm the quality, relevant risk communication and possible removal from the market. This must be conducted in conjunction with clear, tailored and timely communication to the public.

The role of RCCE in the context of substandard and falsified medicines

Informed, engaged and empowered communities are the bedrock of successful readiness and response for emergencies, including incidents involving substandard and falsified medicines. The principles of RCCE are outlined in the [10 steps to community readiness](#) package (3).

The desired outcome of effective RCCE is to mitigate the potential negative impact of health hazards before, during and after public health emergencies or unusual events, working with the health sector and beyond (4). The ultimate goal of RCCE during health emergencies is to reduce morbidity and mortality by empowering communities to confidently participate in leadership, planning and implementation of activities throughout the health emergency response cycle. This is the reason why risk communication is one of the core technical capacities under the International Health Regulations (IHR) (2005) (5, 6) and should be an integral part of incident management support teams (IMSTs) globally.

Effective RCCE should result in affected communities and target audiences knowing how to protect themselves and others against a health threat or emergency, how to seek care, testing or treatment, and how to prevent, manage and avoid stigma and discrimination. To achieve this, communities at risk need to be included and consulted in developing strategies and plans and in the implementation of readiness and response activities to emergencies (7).

The aim of RCCE activities in the context of substandard and falsified medicines is to inform and educate both the public and providers of medical products about the potential threat that substandard and falsified medicines pose to health and empower them to adopt safer practices (8).

In the readiness phase, education and awareness are vital to ensuring patients and consumers have the necessary information on the risks posed by substandard and falsified medicines, how to detect and avoid them, what steps to take if they believe they are exposed, and more. It is also crucial that national policies, strategies, processes and procedures are established to aid in incident management, ensuring accurate, timely and balanced risk and crisis communication when relevant.

During the response phase, the focus of RCCE shifts to reducing risks to patients and consumers, securing supply chains, addressing uncertainty, maintaining ongoing education and strengthening collaboration with relevant stakeholders to improve the response (9).

WHO response to incidents involving substandard and falsified medicines

Substandard and falsified medicines are produced, distributed and sold worldwide, representing a long-standing and ongoing global concern. These medicines can increase disease prevalence, worsen antimicrobial resistance and cause adverse health effects, while also wasting resources. This can lead to economic losses, higher out-of-pocket medical expenses and damage trust in national institutions and regulatory authorities responsible for ensuring the quality, safety and efficacy of medicines within their countries. Preventing, detecting and responding to incidents involving substandard and falsified medicines remain persistent public health challenges for WHO Member States (10).

WHO works constantly with regulatory authorities, manufacturers and other partners, including the public, to prevent, detect and respond to incidents involving substandard and falsified medicines. WHO issues global medical product alerts, urging action to protect communities from substandard and falsified medicines, and works with countries to promote securing the supply chains of the raw materials used to manufacture medical products.

In 2012, the World Health Assembly established the Member State Mechanism, a global platform where countries can convene, coordinate, decide and organize actions to address substandard and falsified medical products. This platform promotes effective collaboration between Member States and WHO in the prevention, detection and response to substandard and falsified medicines and medical products to save lives (11). It was

established to protect public health, prevent incidents involving substandard and falsified medicines and promote access to affordable, safe, efficacious and quality medical products (1).

In 2013, WHO launched the Global Surveillance and Monitoring System (GSMS) portal to strengthen global coordination against substandard and falsified medical products. The GSMS is not a stand-alone system – it is integrated with national reporting mechanisms, enabling a bidirectional flow of information between WHO and Member States. Countries report incidents in a structured and systematic manner, contributing to a more accurate understanding of the scope, scale and harm caused by substandard and falsified medical products. In turn, WHO provides timely alerts, technical support during emergencies and facilitates cross-border linkages between incidents. This collaborative model enhances both national and global capacity to detect, assess and respond to substandard and falsified threats effectively.

WHO continues to assist Member States in strengthening their regulatory ability to combat substandard and falsified medical goods (9), publishing recommendations on good governance and transparency, particularly in the procurement of medical products. Poor procurement practices have resulted in substandard and falsified drugs reaching patients, as have medicine shortages, which are exploited by individuals involved in the manufacture, distribution and supply of substandard and falsified medicines. Anticipating and preventing shortages, together with transparent procurement methods, can help prevent substandard and falsified medications from entering supply chains and reaching patients.

Purpose of the toolkit

The purpose of this toolkit is to guide RCCE practitioners, decision-makers, including regulatory authorities for medical products, and partners on how to place affected communities at the centre of coordinated efforts to reduce the impact of incidents involving substandard and falsified medicines that pose a public health risk. It provides strategies, best practices and practical resources to collect and analyse social and behavioural data; use collected insights to inform strategy and implementation; coordinate activities with partners and stakeholders; support the development and dissemination of accurate information to those at risk; address public concerns; and support the participation of communities as essential partners in readiness and response efforts. These principles are vital for more tailored, equitable and inclusive health emergency programmes.

Intended audience

This RCCE readiness toolkit has been designed for use by:

- national and local regulatory authorities;
- RCCE practitioners;
- decision- and policy-makers;
- manufacturers, suppliers and distributors of medicines;
- emergency management authorities;
- law enforcement;
- academic and research institutions;
- United Nations (UN) agencies and other international nongovernmental organizations;
- nongovernmental organizations (NGOs) and civil society organizations (CSOs); and
- community leaders.

How to use the toolkit

The toolkit supports coordinated, inclusive and tailored RCCE, highlighting approaches that are essential for the successful management of incidents involving substandard and falsified medicines, which could be harmful or life threatening. All tools require contextualization based on local social-behavioural data, available partners, capacity, community-specific needs and the status of emergency readiness and response activities. The resources in this toolkit can be used individually or in combination, as appropriate to the needs and context of the emergency or situation. Tools are presented here in a specific order but do not need to be used in sequence.

Good communication around these incidents is not just reactionary: systems and tools should be set in place before and after crises. Poor management of incidents leads to difficulty in explaining delays and the potential for increased public harm. Poor communication can lead to a more serious health event than the initial incident that was reported. Both lead to a loss of confidence in medicines, health systems and authorities, health care professionals and organizational reputational damage.

Those interested in using these tools should coordinate to adapt them for their context using the following steps.

Step 1: Review all tools

This toolkit contains a range of tools with different aims and objectives. It can be used like a library of resources to meet existing country-level needs – not all tools will be relevant or necessary for all settings. All tools should be reviewed and selected for use based on needs and priorities outlined in national plans and strategies for dealing with substandard and falsified medicines. Also, adequate training for end users is essential to maximize effectiveness. Operational and contextual barriers should be assessed and addressed, with continuous updates to the tools, informed by new evidence and field feedback.

Step 2: Adapt the relevant tools

This toolkit has been developed at a global level. All provided resources should be adapted to local contexts. This can be done by national decision-makers, RCCE practitioners or partners and in line with communities engaged in the response. Adaptations that may be needed include the following:

- **Language and audience:** Ensure the tools are translated into local languages and dialects, with careful consideration of literacy and accessibility requirements.
- **User:** Adapt and refine the tools according to the needs of those who will be using them. Different stakeholders have different needs and capacities.
- **Incident context:** Adapt the tools based on the current situation as well as cultural and social norms, local beliefs and practices to enhance acceptance and effectiveness of the designed interventions among the impacted populations. Future adaptations may be needed as the situation evolves. RCCE activities should be conducted in coordination with other emergency response pillars such as surveillance, clinical management for treatment and case management, and others.
- **Phase of the emergency:** How the tools in the toolkit are adapted and implemented will depend on the current phase of the health emergency cycle in the local context. Tool 7 (RCCE readiness and response checklist for incidents involving substandard and falsified medicines) can be used to identify different priorities within the different phases.
- **Existing national activities:** Selection and adaptation of tools should be guided by national action plans, legislation, strategies and ongoing activities to complement and enhance existing efforts. Tools should also be adapted to the level of available technology and infrastructure (e.g. internet connectivity, mobile phone penetration, electricity reliability).

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Step 3: Use and monitor

Once the tools are tailored to the local context, they can be used to inform strategy, planning and guide the implementation of RCCE activities. The resources within the toolkit are designed to support WHO's work but are also valuable for other engaged partners and stakeholders, including national health and regulatory authorities, community leaders and local actors, to enhance their activities. The use of these tools should be continuously monitored and evaluated to inform ongoing improvements.

2. Background information on substandard and falsified medicines

The background information included in this tool is up to date as of November 2025. It is intended to provide RCCE decision-makers, practitioners and partners with the knowledge and understanding needed to respond to public health incidents involving substandard and falsified medicines. Up-to-date information about the local situation should be sought from local incident response leads to establish a full understanding of the local setting.

Overview

The WHO and UN Office on Drugs and Crime (UNODC) landmark report released in July 2025, [Contaminated medicines and integrity of the pharmaceutical excipients supply chain](#), highlights the growing threat posed by contaminated and falsified medicines, calling it a major but preventable cause of patient harm globally (12). It provides evidence of widespread incidents across all regions, with a disproportionate impact on low- and middle-income countries (LMICs). Up to 2 billion people worldwide lack access to essential medicines, including vaccines and other vital health products. This gap between supply and demand is often filled by substandard and falsified medicines and medical products. Up to one in ten medicines sold globally in LMICs may be substandard or falsified, including ones that could be harmful to humans (13). The problem of substandard and falsified medicines is intensifying and becoming more urgent to address as global supply chains grow more complex and distribution and consumption become more globalized, including through online sources. This presents a major challenge for Member States striving to achieve increased access to quality, safe, effective and reliable medicines (14).

- **Substandard medical products** are those that are authorized but fail to meet their approved quality standards or specifications. This may be because they were poorly manufactured, badly packaged or transported. With **falsified medicines**, the manufacturing and packaging are deliberately designed to misrepresent their identity, composition or source. This may be due to

poor manufacturing, inadequate packaging or improper transport. In the case of falsified medicines, the manufacturing and packaging are deliberately designed to misrepresent their identity, composition or source, and the active ingredient levels may be either dangerously high or ineffectively low.

- **Unregistered/unlicensed medical products** are products that have not undergone evaluation and/or approval by the national or regional regulatory authority for the market in which they are marketed/distributed or used, subject to permitted conditions under national or regional regulation and legislation.
- **Contaminated medicines**, a category of substandard medicines, are medicines that are unsafe for use or ineffective because they contain harmful foreign substances or microorganisms, including some chemical, biological or radiological substance or matter that should not be present according to the product's specifications. This can lead to severe sickness, poisoning, infections, treatment failure, drug resistance or even death (15). Contamination can occur when medicines are produced in inadequately controlled settings by poorly trained personnel, and/or shipped or accidentally stored without observing requirements and standards. On occasion, medicines are contaminated on purpose, whether to save money or because an ingredient is hard to find or buy (15).

- Reports received by WHO from Member States suggest that all therapeutic categories are affected by substandard and falsified medicines, particularly anti-infectives, antimalarials, cardiovascular, oncology products and pain medications (16). It is estimated that 72 000–169 000 deaths are caused annually by substandard and falsified antibiotics in children under 5 suffering from pneumonia, and another estimated 31 000–116 000 deaths annually caused by substandard and falsified antimalarials in sub-Saharan Africa (9). Recently, several countries in Africa, Asia Pacific and Europe have identified and reported medicines that have been contaminated and are dangerous to human health. Between October 2022 and April 2023, a series of contaminated medicines (such as cough syrups) were reported in Cambodia, The Gambia, Indonesia, Marshall Islands, Micronesia (Federated States of) and Uzbekistan, and more than 300 people, predominantly children, reportedly died after having potentially taken one of these contaminated products (15).

These irregularities can undermine people's confidence in medical systems and endanger health, while having a huge impact on individuals, families, national health systems and the economy (13). RCCE is essential to preventing panic, maintaining trust and ensuring that information is available on safe alternatives during recalls.

How substandard and falsified medicines enter the market

Substandard and falsified medicines typically enter the market due to weak regulatory systems, inadequate quality control, and poor supply chain management. These products often bypass standard regulatory checks and are distributed through unregulated or poorly regulated channels. This is more likely to occur when people require or want medicines that they cannot access or afford, or when there are shortages further up the supply chain, as pharmacies, hospitals and other health and care providers frequently fail to meet their patients' needs or demands.

Sometimes, shortages arise on a larger scale if a necessary product is too expensive for the national health system, or in the context of an emergency, because an unexpected disease outbreak has resulted in demand beyond the global production capacity of known suppliers. In all such cases, people and institutions do things they would not normally do, for example, order a batch of medical supplies from an unknown, untrusted and unlicensed source (13). The WHO/UNODC report emphasizes that contaminated medicines are not limited to substandard

manufacturing but also involve criminal networks trafficking falsified products (12).

Where substandard and falsified medicines are found

Substandard and falsified medicines are frequently found in informal, unlicensed and illegal outlets, including street markets, hawkers and illegal online sales. However, they are also sometimes reported from legal and regulated sources such as hospitals, clinics, pharmacies and drug stores in all regions (16).

Substandard and falsified medicines are most likely to be found where:

- access to available, affordable, quality, safe and effective medicines is constrained;
- standards of governance are low, from poor ethical practices in health care facilities and medicine outlets to inefficient administrative systems and corruption in both the public and private sectors; and
- the tools and technical capacity to ensure good practices in manufacturing, quality control and distribution are limited, and there is weak oversight of supply chain security or insufficient reporting mechanisms (1).

Substandard and falsified medicines and the internet

Unregulated websites, social media platforms and smartphone applications can also be direct sources of substandard and falsified medicines (see Annex). Unlicensed and unregulated sources significantly increase the risk to consumers, who should be cautious of the following:

- spam email advertising medicines;
- lack of authenticity – no verification logo or certificate;
- spelling mistakes and poor grammar on the packaging, or on websites or advertisements;
- websites that do not display a physical address or landline;
- websites offering prescription-only medicines without a prescription;
- suspiciously low-priced products; and
- addictive substances and fancy products (e.g. Viagra) are more likely to be advertised on the internet, having the chance of market penetration through online sources.

The dangers of substandard and falsified medicines

Substandard and falsified medicines can:

- endanger health, prolong illness and even kill;
- promote antimicrobial resistance and the spread of drug-resistant infections;
- undermine confidence in health professionals and health systems;
- create distrust about the effectiveness of vaccines and medicines;
- eat into the limited budgets of families and health systems;
- provide income to criminal networks (13);
- distort pharmaceutical markets and discourage genuine manufacturers; and
- create cross-border public health emergencies.

Who is at risk

Low- and middle-income countries and nations facing armed conflict or civil unrest, where health systems are fragile or non-existent, bear the highest burden of substandard and falsified medicines (1). The same applies to countries experiencing emergencies created by natural disasters, such as earthquakes, hurricanes or catastrophic flooding, which increase the demand for health services and medicines while also disrupting their delivery. Overall, humanitarian settings are high risk due to disrupted regulatory oversight.

The issue of substandard and falsified medicines is not limited to low-income countries; with the increase in internet use, access to substandard and falsified products has also increased, with those manufacturing, distributing and supplying these medicines able to access a global marketplace more easily. An emerging global market for so-called lifestyle products – such as erectile dysfunction treatments (which may be therapeutic for medical conditions but are often classified as lifestyle medicines when used to enhance quality of life), hair loss remedies, weight loss medicines and cosmetic interventions – combined with a growing culture of self-diagnosis and self-prescribing, has resulted in thousands of unregulated websites offering unsupervised access to substandard and falsified medicines.

Prevention and readiness

At a national level, preventing incidents involving substandard and falsified medicines requires legislation and regulations to stop the manufacture, sale and consumption of substandard and falsified medicines and ensure they do not enter supply chains and reach patients and consumers. A mature regulatory system minimizes the risk of substandard and falsified products infiltrating the regulated and formal supply chain through effective prevention, detection and response strategies. It is also vital for governments to register medicines and carry out systematic inspections of manufacturing and distribution facilities to ensure compliance with national and internationally recognized standards.

A clear legislative framework that outlines requirements for compliance and includes dissuasive sanctions for non-compliance is important. The goal is to create an environment where manufacturers and wholesalers feel confident to promptly report honest mistakes, knowing that the regulatory approach will be measured, consistent and proportionate (13). To assist Member States, WHO issued guidance and assessment tools to help prevent substandard and falsified medicines from entering supply chains and reaching patients (9). At the individual or community level, the following steps will help minimize the threat posed by substandard and falsified medicines if they have entered the supply chain:

- Source medicines only from regulated or licensed outlets, e.g. pharmacies, clinics, doctors, hospitals, health centres.
- Avoid unauthorized websites supplying medicines, especially those that conceal their physical address or landline telephone number.
- Examine the product packaging for its condition, spelling mistakes or grammatical errors.
- Inspect medicines before consumption and ensure the medicine looks correct, is not discoloured, degraded or has an unusual smell.
- If there is suspicion that it is not working properly or you have suffered an adverse reaction to a medicine, discuss with a pharmacist or doctor as soon as possible (15).
- Check the price of medicines: be cautious of prices that are significantly lower than usual, as this may indicate falsified products. Avoid purchasing medicines with suspiciously low prices.
- Always follow medical advice: use only medicines prescribed by health care professionals and avoid self-medication, especially for serious illnesses.

3. Tools to support readiness and response activities involving incidents of substandard and falsified medicines

Gathering social-behavioural information and data

The tools in this section are designed to support the collection, analysis and use of social-behavioural data and community insights to inform the development of evidence-based RCCE strategies and plans. The data and insights collected using these tools promote better decision-making and can allow for stronger risk assessments by bringing a community lens to the understanding of risk during an emergency or incident involving substandard and falsified medicines. By prioritizing the collection, analysis and use of social-behavioural data and community insights within and beyond RCCE, it is possible to bring broader response strategies and plans in line with community expectations, needs and priorities.

Tool 1: Audience analysis

This tool should be used to identify and record key information about key audiences and communities involved in substandard and falsified incidents and who should be included in readiness and response activities. The collected information should be used to inform and select the appropriate RCCE strategy and action plans for the priority communities at risk.

In order to have inclusive and comprehensive RCCE plans and strategies, it is crucial to involve relevant audiences in co-designing solutions and interventions aimed at preventing and protecting against threats posed by incidents involving substandard and falsified medicines. Understanding each audience, their role and their needs helps identify who is most at risk, how they can be impacted and how to support communication and engagement efforts. Encouraging active community participation in detecting and reporting substandard and falsified medicines can significantly enhance the effectiveness of readiness and response actions.

In the case of these incidents, it is imperative to ensure that all RCCE activities focus on equitability. There is a risk that RCCE activities can

increase health inequities if they end up reaching and benefiting already-privileged groups rather than less-privileged groups who are more at risk of the dangers posed by substandard and falsified medicines. Users of the toolkit should also assess potential impacts on the population according to the findings of the situational analysis (PESTEL – political, economic, sociological, technological, legal and environmental) (Tool 3), behavioural analysis (Tool 4) and risk analysis (Tool 5).

Table 1 provides an audience analysis matrix, which helps to collect and organize information about key audiences and, in combination with other tools in the document, provides a broader context to help tailor RCCE activities in support of equitability, healthy behaviours and the requirements of the specific communities.

Table 1. Audience analysis matrix

	Patients and the general public	Health care professionals (doctors, pharmacists, nurses...)	Health and regulatory authorities (ministries of health, NRAs, national institutes of health...)	Industry (pharmaceutical companies, manufacturers, wholesalers, distributors...)	Media
Demographic information – age, gender, languages spoken, literacy levels, education, employment status, urban vs rural, etc.					
Risk level – based on findings from risk assessment. Can be ranked low, medium or high, or based on risk assessment scale.					
Perceived level of risk – based on the situational and behavioural analysis and the level of information available related to the substandard and falsified medicine in circulation. Can be ranked as low, medium or high.					
Audience attitudes – trust in health and regulatory systems, concerns over the impact of substandard and falsified medicines, fear and mistrust regarding medicines, previous experience with substandard and falsified incidents.					
Trusted information channels – journals, conferences, official reports, social media, public health announcements and awareness campaigns, safety bulletins, interviews.					
Audience needs and concerns – need for accurate, timely information about the substandard and falsified incident and safety, guidance and resources, data, collaboration.					
Influential voices – celebrities, thought leaders, health workers, social media accounts, etc.					
Access to key interventions – testing, treatment, etc.					
Rumours and misinformation circulating.					

Tool 2: Stakeholder analysis

This tool looks at the various people and groups identified as important in substandard and falsified incidents and response activities, or impacted by the incident. It helps to bracket and group their potential roles, capacities and anticipated engagement to support collective efforts to prevent or respond to incidents involving substandard and falsified medicines.

A stakeholder analysis goes into more detail and builds on the findings of the audience analysis (Tool 1), situational analysis (PESTEL) (Tool 3) and behavioural analysis (Tool 4). This tool should be adapted to the local context to provide a precise overview of different stakeholder roles, motivations, anticipated involvement and key milestones to maximize the impact of RCCE activities.

There are four main categories into which stakeholders fall, with an associated strategy for interacting with them, as seen in Table 2. In the context of substandard and falsified medicines, relevant stakeholders expand to include the NMRA, health care professionals, procurement agencies, manufacturers, importers, wholesalers, distributors, retailers, customs and police, and more.

Table 2. Stakeholder categories

	Stakeholder category	Strategy
Champion	Champions support your activities and do so actively and visibly. These groups/people agree with the proposed actions and goals and are already taking action on their own to support them, i.e. UN, other state agencies or authorities, etc.	With champions, continue engaging them in planning and implementation of activities, provide them with updates and information to ensure they are up to date, appreciate and acknowledge their contributions and support, and let them champion the cause.
Silent booster	Silent boosters support the planned or proposed activities and goals but do so privately, with little to no public support. These stakeholders need additional motivation to become more active and supportive of the proposed actions.	With this group, the strategy is to educate, enable, inform and motivate. Energize these stakeholders by involving partners and champions they respect and normally engage with to help advocate for the planned activities and goals.
Avoider	Avoiders don't necessarily support your cause but aren't vocal or visible about their lack of support. They silently oppose aspects of planned activities and passively disagree.	Inform or ignore. With avoiders, it is helpful to engage groups from the champions category to help influence them to support activities.
Blocker	Blockers are groups who are visibly, publicly opposed to the planned activities and take action to encourage others to disagree as well. They pose an obstacle to the implementation of activities, depending on their influence.	Blockers pose a greater challenge if they are influential. If they are, the best approach is to counteract their action by continuing to enlist champions to advocate for your cause and provide facts. If they are not influential, the best strategy is to ignore this group. Regardless, keep track of who they are and who they are influencing.

Tool 3: Conducting a situational analysis: the PESTEL tool

A situational analysis can be conducted in either the readiness or response phase to inform activities. In any of these scenarios, the situational analysis should be regularly updated.

The PESTEL tool is a framework for conducting a situational analysis that helps understand political, economic, sociological, technological, environmental and legal factors that can influence public health efforts during an incident involving substandard and falsified medicines, including non-reporting.

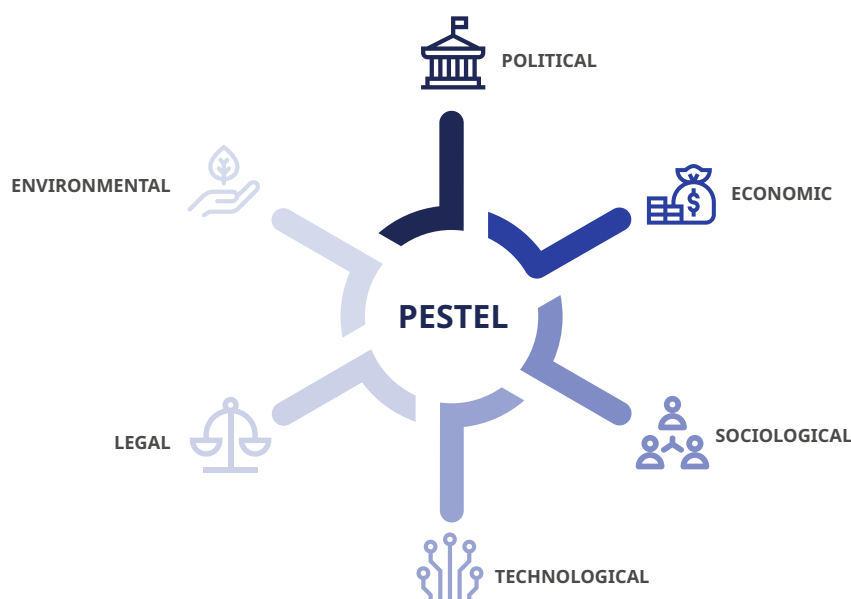
Data collected either directly or from existing sources can be used to gather insights into the six categories of the PESTEL analysis. Information can be collected through these and other sources:

- community surveys, qualitative interviews and focus group discussions, including behavioural science research;
- tools used under the [International Health Regulations](#) (5) to evaluate country capacity, including intra-action reviews, after-action reviews, the [Health Resources and Services Availability Monitoring System](#) (HeRAMS), [Joint External Evaluation \(JEE\) reports](#) (4), etc.;
- lessons learned from previous emergency responses;
- [WHO Global Surveillance and Monitoring System](#) (17);
- pharmacovigilance and poison centres;

- WHO IMST updates, situation reports, [Disease Outbreak News](#) (18) and daily reports;
- peer-reviewed journals;
- WHO and other reputable country profiles e.g. Centres for Disease Control;
- news reports from trustworthy sources; and
- government and NRA websites and official publications, including legislation.

The information obtained from the PESTEL analysis can be complemented by other data sources and used in combination with detailed behavioural data from Tool 4. This supports the validation of findings and further analysis of relevant insights.

Fig. 1. PESTEL analysis framework



Political considerations

- governmental and local policies;
- budgets for emergency readiness and response;
- policies and strategies for the regulation of medicines, including availability of a substandard and falsified national action plan;
- level of political will to enforce substandard and falsified medicine regulations;
- previous governmental experiences with the enforcement and management of incidents involving substandard and falsified medicines;
- levels of trust in government, specifically regulatory authorities like the NMRA, partners and other influential voices;
- government and partner public communication activities and style;
- upcoming elections or potential changes in leadership; and
- source of the medicines (i.e. which country they came from) and the relationship between the impacted countries.

Economic considerations

- capacity of citizens and communities to participate in economic life – economic participation enhances access to resources, education and health care, which are crucial for effective emergency preparedness and resilience during crises;
- access to and supply of health services, including well-regulated medicine supply chains and/or informal markets;
- income and out-of-pocket expenditure of citizens;
- the economic value of the affected medicines; and
- financial impact on the affected manufacturer and the national economy.

Sociological considerations

- cultural dynamics and demographics;
- societal trust in national health systems, including regulatory authorities;
- behaviours, beliefs and habits;
- religions and tradition;
- stigma or discrimination of marginalized groups; and
- literacy, languages and dialects.

Technological considerations

- level of access to information (print, broadcast or online media);
- mobile phone usage and level of penetration;
- social media usage;
- availability of internet access;
- use or availability of detection technologies and traceability systems;
- digital literacy; and
- key online communication channels.

Environmental considerations

- environmental risks from the inappropriate disposal of unwanted, including substandard and falsified, medicines; and
- environmental risks from manufacturing medicines.

Legal considerations

- legislative and regulatory frameworks for the manufacture, storage, procurement and distribution of medicines;
- applicable international treaties, e.g. World Trade Organization agreement on Trade-Related Aspects of Intellectual Property Rights;
- laws, rules and plans, including those related to ethics, such as the prevention of sexual exploitation, abuse and harassment;
- existence of treaties or binding legal instruments;
- multiple levels of governance;
- regulations that impact RCCE in emergency situations; and
- coordination and engagement of national pharmacovigilance reporting centres, NMRAs, national poison centres and national quality control laboratories.

Tool 4: Behavioural analysis

This tool can be used to identify and understand behaviours relevant to incidents involving the safety and efficacy of substandard and falsified medicines, thereby informing and shaping RCCE strategies, tools and tactics. During a health emergency, behaviours are dynamic rather than static; high-risk behaviours can be influenced by a range of barriers and enablers, which can be identified through the collection of social and behavioural data. This information should be gathered as early as the prevention phase and monitored regularly throughout the readiness and response phases to understand norms, trends and behavioural changes (19).

Used together, the findings from the situational and behavioural analyses can help assess how to engage with communities effectively and co-develop plans and strategies that support people to make well-informed decisions to protect themselves. The importance of including behavioural and social sciences in public health interventions was globally recognized in 2023 by Member States at the [Seventy-sixth World Health Assembly](#) (20), where WHO acknowledged the contribution of these disciplines in achieving improved health outcomes and called on the increased use of behavioural science to empower communities in understanding public health problems and designing and evaluating interventions to address them.

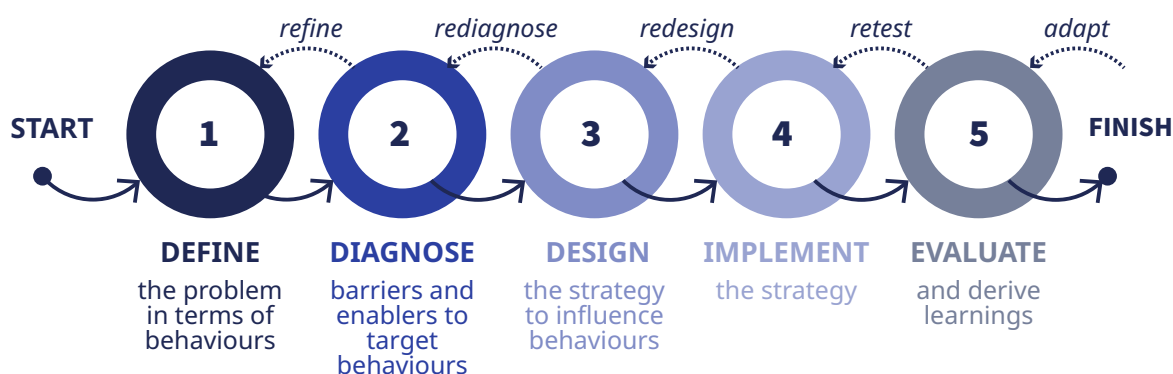
Behaviours are shaped by the interplay of cultural, social, economic and environmental factors and can influence the uptake of protective actions and care-seeking practices in health emergencies. It is essential to identify and understand risky and protective behaviours in the context of the substandard and falsified incident, as these insights can inform the development of RCCE strategies, plans and activities. Importantly, while changing behaviour is a critical component

of addressing an emergency, it is not the sole solution; individuals require access to information, opportunities for engagement and resources for prevention and care. This support empowers them to make informed decisions that are relevant to their daily lives and are both practical and accessible.

The behaviours that are relevant to exposure to substandard and falsified medicines will vary depending on the local context. This information should be obtained from a multidisciplinary team, including behaviour change experts working on the response and from the PESTEL analysis.

The behavioural insights checklist based on the define, diagnose, design, implement and evaluate (DDDIE) steps (Fig. 2) is designed to guide which data to review to inform the RCCE strategy, including inputs from the communities at risk. It is adapted from the [Technical note from the WHO Technical Advisory Group on behavioural insights and science for health](#) (21). This technical note gives additional guidance on behavioural insights including advice on the principles and application of behavioural science.

Fig. 2. DDDIE steps guide



Step 1: Define the problem in terms of behaviour?

Are there any specific behaviours that are contributing to substandard and falsified incidents?

Use the data sources available to answer the following questions and complete Table 4 (e.g. existing social-behavioural data, knowledge from previous local incidents or incidents in other countries).

1. Does the problem have a behavioural component? Consider factors such as:

- What is driving the use of substandard and falsified medicines?
- Are people practising protective behaviours?
- Are people practising risky behaviours?

2. Which behaviour(s) must be changed to contribute to improve or attain the desired health outcome(s)?

3. What is the target behaviour you are aiming for? Specify who needs to do what, when, where and how. Be as specific as possible, while recognizing that behaviours are interconnected. The target behaviour is likely part of a sequence involving multiple key players, occurring at different times and places, and all contributing to transmission.

Problem example: The issue involves the purchase and use of medicines that are marketed as traditional, herbal or natural but are actually substandard or falsified by the addition of unclaimed substances, including conventional medicines or prohibited substances. This represents the following health risks:

- Unpredictable dosages: the amount of undeclared substances is unknown, leading to potential overdose or adverse reactions.
- Drug interactions: undeclared substances can interact negatively with other medications or existing health conditions.
- Lack of quality control: substandard and falsified medicines are not subject to the same rigorous quality control as regulated pharmaceuticals.
- Misleading claims: consumers are deceived into believing they are using safe, natural remedies, undermining informed consent.
- Delayed appropriate treatment: people who rely on these medicines may delay seeking needed medical care.

Table 4. Problem and behaviour diagnosis

Step 1: Defining the problem in terms of behaviour	
Does the problem have a behavioural component? If yes, what?	Yes, the problem has a significant behavioural component. • Consumer behaviour: seeking and purchasing unregulated traditional medicines. • Producer/distributor behaviour: intentionally adulterating and misrepresenting medicines.
Which behaviour(s) must be changed to improve the desired health outcome?	• Consumers' reliance on unregulated traditional medicine sources. • Producers' and distributors' fraudulent practices. • Consumers not reporting adverse drug reactions. • Consumers not seeking advice from qualified health care professionals.
What is the target behaviour you are aiming for?	• Purchasing from regulated sources: buy medicines only from licensed and authorized sellers. • Verification of products: check for proper labelling, certification and regulatory approval of the medicines. • Consulting health care professionals: always consult with health care providers before using any new medicines. • Producers and distributors adhere to regulatory standards and ethical practices.
Who needs to change their behaviour?	• Consumers: individuals who purchase and use unregulated traditional medicines. • Vendors: sellers who distribute these products need to ensure they are selling legitimate and safe medicines. • Regulatory bodies to enforce appropriate controls.

Table 4. Problem and behaviour diagnosis (cont.)

Step 1: Defining the problem in terms of behaviour	
What do they need to do differently?	• Consumers: » Purchase medicines only from reputable and licensed sources. » Verify the authenticity and quality of the medicines according to the latest available guidance. » Consult health care professionals before using new the medicines. » Increase their health literacy. • Vendors: » Ensure that all medicines sold are properly labelled and certified. » Avoid selling unregulated or uncertified medicinal products. » Avoid adulteration and misrepresentation.
When does this behaviour occur?	• When individuals experience health concerns and seek remedies. • When individuals seek cheaper alternatives to prescribed medications. • When individuals believe traditional or natural remedies are safer. • When individuals are influenced by misleading advertising.
Where does this behaviour occur?	• Consumers: in informal markets, online and social media platforms and unlicensed pharmacies. • Vendors: at points of sale, including street markets, online stores and unlicensed pharmacies. • In communities with limited access to health care and education. • In areas with weak regulatory oversight.

Step 2: Diagnose the barriers to and enablers of target behaviours

A barrier is an obstacle or challenge that impedes the uptake or adherence to interventions related to substandard and falsified medicines. In contrast, enablers are factors that facilitate or support the successful implementation of preventive measures and RCCE interventions. Barriers and enablers of behaviours can be cognitive/psychological; social/cultural; environmental/structural and linked to perceptions of self-efficacy, risk and efficacy of interventions, as well as other factors. In the context of substandard and falsified medicines, some key barriers may be a lack of awareness, the absence of a system or method for reporting, overcomplicated reporting procedures, limited response from regulatory authorities or a lack of feedback.

Identifying and understanding the barriers and enablers of the desired target behaviour is essential to designing public health interventions that are effective, practical and culturally appropriate. Apply insights from social and behavioural science, along with relevant evidence, to prioritize and select the key barriers and enablers to explore further. This process will inform the development of interventions that are

both impactful and context-sensitive. An example can be seen in Table 5.

It can be useful to consider whether barriers and enablers are: 1) cognitive/psychological; 2) social/cultural; 3) environmental/structural.

Examples of barriers

- Lack of awareness or knowledge about substandard and falsified medicines, scope, scale, harm and risks and access (cognitive/psychological).
- Lack of trust in health or regulatory authorities (cognitive/psychological).
- Cultural beliefs and practices that contradict guidance or discourage people from adopting the desired behaviours (social/cultural).
- Limited access to medicines, health care resources or services required to follow the desired behaviours (environmental/structural).

Examples of enablers

- Strong community inclusion, support and engagement in promoting interventions, including reporting of contaminated medicines (social/cultural).
- Accurate RCCE interventions that provide information about substandard and falsified medicines (environmental/structural).
- Positive social norms that encourage and support people in adopting desired and avoiding risky behaviours, as well as encouraging reporting of incidents without reprisal (social/cultural).
- Engaging positive role models, such as community leaders and influencers, to advocate for and model the desired behaviours (social/cultural).
- Accessible and reliable health services to support the adoption of the desired behaviours (environmental/structural).

Table 5. Behaviour barriers and enablers

Step 2: Diagnosing the barriers to and enablers of target behaviours		
Risky behaviour examples	Enablers examples	Barriers examples
Seeking and purchasing unregulated traditional medicines.	Social or cultural preference for traditional medicines.	Inability to access or afford prescriptions and authorized medicines.
Buying medicines from street vendors or unlicensed sellers.	Lower cost and easier access. Lack of awareness about risks.	Weak regulatory enforcement.
Purchasing from unverified or illegal websites.	Self-diagnosis and self-medication. Convenience and anonymity.	Lack of legal action and enforcement against illegal online sellers.
Procurement of medicines through corrupt channels which bypass regulatory standards.	Weak institutions and lack of accountability; limited transparency in procurement and supply chains.	Poor governance and lack of anti-corruption frameworks.

Steps 3, 4 and 5: Design, implement and evaluate interventions to address barriers and encourage enablers of target behaviours

Steps 1 (define) and 2 (diagnose) provide insights and data that can then be used in steps 3 (design of evidence-based RCCE approaches and interventions aimed at addressing the barriers identified), 4 (implementation of interventions aimed at addressing the barriers identified) and 5 (evaluation) to support readiness and response efforts around incidents involving contaminated medicines.

Design and implementation of interventions should be done in collaboration with behavioural scientists, health experts, communication specialists and, crucially, with affected communities and stakeholders, ensuring the design of effective and culturally sensitive interventions. Tools to support implementation are included in this toolkit (Section 3: Implementation).

Evaluation of interventions and behaviour change is important to drive future learning about the effectiveness of RCCE strategies. It is possible to measure the impact of interventions on behavioural outcomes using epidemiological data or direct observations of behaviours. If these data are not available, use self-reported information, such as adherence to preventive measures (when available). It can also be useful to include process evaluation indicators to understand how context, implementation and mechanisms of impact may have influenced outcomes.

Tool 5: Risk assessment for incidents involving substandard and falsified medicines

A risk assessment can be conducted in either the readiness or response phase to inform activities and to support Member States in identifying high-risk incidents involving substandard and falsified medicines. Using the WHO risk assessment tools below alongside other tools such as Tool 3 (PESTEL) and Tool 4 (Behavioural analysis) is essential in developing RCCE strategies and interventions.

The goal of a risk assessment is to identify the magnitude of a hazard or risk to the public and to prioritize actions that will mitigate the threat. This requires systematically gathering, assessing and documenting information to assign a level of risk, which provides the basis for taking action to manage and reduce the negative consequences of substandard and falsified medical products. Use of a risk assessment tool, such as the [Rapid risk assessment of acute public health events: manual](#) (22), in conjunction with Table 6 indicating substandard and falsified incident priority levels and criteria, is especially useful for the analysis of cases involving falsified, unregistered or unlicensed and substandard medicines. The document [Recommendations for health authorities on criteria for risk assessment and prioritization of cases of unregistered/unlicensed, substandard and falsified medical products](#) (23) also provides further

details on the rationale and process of a risk assessment specific to substandard and falsified medicines.

The risk management cycle includes the identification and prioritization of mitigation measures, taking into account the likelihood of success, feasibility of implementation and unintended consequences. Multistakeholder engagement is always encouraged to establish all necessary facts. Risk assessment is a continuous process and requires continuous monitoring and evaluation as the incident unfolds.

Handling incidents involving substandard and falsified medicines in a professional, consistent, timely and proportionate manner requires sound risk assessment and will reflect well upon the organization(s) involved (24).

Table 6. Prioritization and risk parameters by type

Public health	High	- Fatalities or serious adverse reactions, or - Immediate or likely threat of fatalities or serious adverse reactions, or - Multiple numbers of persons affected > 500
	Medium	- Severe adverse reactions, or - Threat of severe adverse reactions, or - Multiple numbers of persons affected > 100
	Low	- Threat of severe adverse reactions, and - Low numbers of persons affected < 100, or - Isolated incident
Supply chain	High	- Product has entered the regulated/licensed supply chain, and - Reached hospitals, clinics, pharmacies or patients, or - Recall is likely to lead to a significant shortage
	Medium	Product is available only through unlicensed and unregulated outlets
	Low	Product not thought to be available to the public
Geographic	High	Product available on an interregional or global basis, including unregulated websites
	Medium	Product restricted to one region
	Low	Product restricted to small area

Environmental	High	Serious or immediate risk to the environment
	Medium	Potential risks to the environment
	Low	No environmental risks identified
Public interest	High	- Widespread and escalating international media interest, or - Patient group or other major stakeholder interest, or - Risk of serious reputational damage to public authorities or organization
	Medium	- Regional or national media interest, or - Subject of complaints to/from stakeholders
	Low	No significant public interest
Political	High	Subject of significant national government interest
	Medium	Affects or is likely to affect, question or change policy
	Low	Non-contentious issue
Economic	High	Major financial gain or serious loss > US\$ 1 million
	Medium	Financial gain or loss < US\$ 1 million
	Low	No significant gain or loss
Legal	High	Exposes your organization to litigation or other legal challenge
	Medium	Exposes weakness in regulatory or legislative regime
	Low	No perceived legal risks
Product history	High	Same batch/lot of product previously notified
	Medium	Same product previously notified
	Low	No previous reports
Supply chain elements	High	Same importer, exporter, broker, wholesaler, distributor or retailer
	Medium	Similar method of transportation, concealment or payment
	Low	No previous reports

Strategy and planning

The tools in this section are designed to support the development of evidence-based RCCE strategies and plans drawing on social-behavioural data, community insights, epidemiological data and priorities identified by other areas of the emergency response. Strong strategies and plans promote more effective implementation of activities in the long run and provide an opportunity to consider how to work with communities as core partners in all RCCE activities.

This tool is designed to determine which communication strategy to apply based on the risk perception of the public and the magnitude of the hazard.

Tool 6: Identifying the appropriate risk communication strategy

Risk perception in public health refers to how individuals think and feel about the health risks they face. It plays a crucial role in determining whether communities and individuals adopt protective behaviours, such as adhering to recommendations from health and regulatory authorities or avoiding substandard and falsified medicines. Understanding how people

perceive health risks, the accuracy of these perceptions, and how individuals receive and interpret information about their own risk is essential (25). Risk perception may vary across various subgroups of stakeholders and targeted messaging is essential once the strategy is selected.

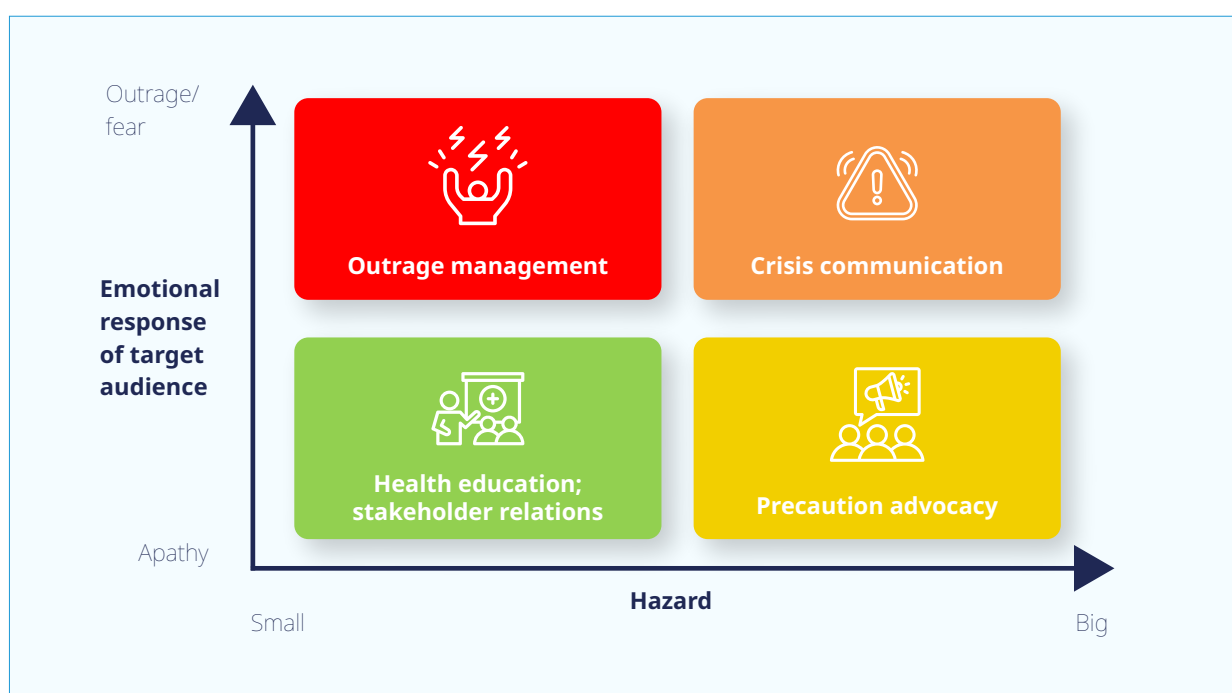
An individual's perception of risk is complex and shaped by multiple factors. It primarily involves two dimensions:

- **Cognitive:** relating to a person's knowledge and understanding of a particular topic and its associated risks.
- **Emotional:** involving feelings, fears and concerns about the risk, as well as apathy.

For technical experts, risk is typically assessed based on the nature and severity of the hazard, such as the clinical or epidemiological consequences of exposure to substandard and falsified medicines. However, for the general public, risk perception is influenced more by emotional responses and the degree of concern or fear the issue generates – often referred to as “outrage”.

The actual scale of the risk, as identified through a risk assessment (Tool 5), combined with public risk perception based on audience, situational and behavioural analyses (Tools 1, 3 and 4) determines the appropriate risk communication strategy. These strategies are from the Sandman model of risk communication and the formula “risk = hazard + outrage”. The model highlights four approaches depending on how big or small the hazard is and where the audience is on a spectrum from apathetic to outraged (26).

Fig. 3. Sandman model of risk communication (risk = hazard + outrage)

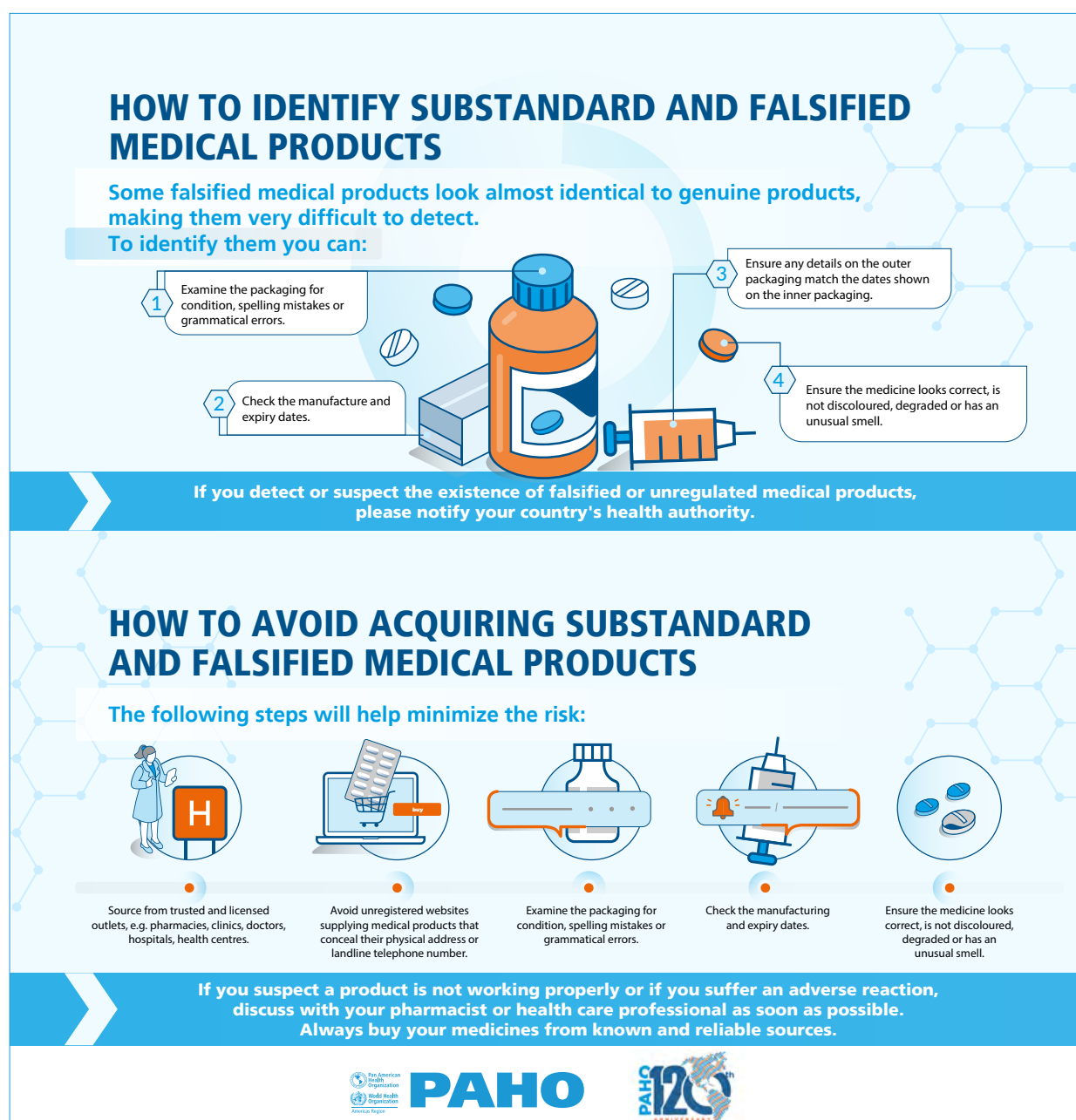


Based on the Sandman model (Fig. 3) and the relevant data collected through the tools provided in this toolkit, you can identify one of four possible strategies to utilize in response to an incident involving substandard and falsified medicines.

When the hazard is relatively small and emotional engagement is low or there is apathy among the audience(s), the appropriate strategy is **health education**. The goal of this approach is to monitor communications surveillance to identify and address an emerging outrage early, before it escalates to outrage management, and to maintain public and stakeholder engagement in ongoing activities or issues. When the risk

is low, maintaining regular communication helps build trust and prepares the public for potential changes in the situation. This strategy remains important throughout the whole risk communication cycle as it provides the opportunity to bridge information gaps and close voids in knowledge to prevent mis- and disinformation later if the outrage or the hazard posed by the incident escalates. The social media tile (Fig. 4) shows an example of health education, informing the general public on how to identify substandard and falsified medicines and avoid taking them (27).

Fig. 4. Examples of health education communication on substandard and falsified medicines



Source: Pan American Health Organization; 2022.

Precaution advocacy is the strategy to be applied when the hazard posed by the incident involving substandard and falsified medicines is significant, but the audience is not very concerned, outraged or even aware of it. This approach aims to raise public concern and motivate people to take action, while carefully avoiding causing too much fear or outrage. Most public health efforts fall into this category. The health issues addressed, including incidents involving substandard and falsified medicines, pose serious threats, but audiences are not emotionally engaged enough to respond or follow public health advice. An example is communicating the dangers of substandard and falsified antibiotics and how their use increases antimicrobial resistance (Fig. 5) (28).

The third possible strategy is **outrage management**, when the hazard is small (little or no real danger) but people are very outraged or upset, or their response is out of proportion to the real risk. When working with this strategy, the focus is on respectfully calming the public down, continuing to communicate facts and evidence while acknowledging any anger or fear among the public and acknowledging their concerns. It is essential to tackle mis- and disinformation, resolve rumours and provide clarity on the actual risks faced by the audience as a result of the emergency or issue. Outrage management is commonly used when public fear is disproportionate to the level of risk or not grounded in science and evidence, as misinformation spreads rapidly, as was the case during the rollout of COVID-19 vaccines (Fig. 6) (29).

Fig. 5. Example of precautionary advocacy communication

Source: Fight the Fakes Alliance; 2025.

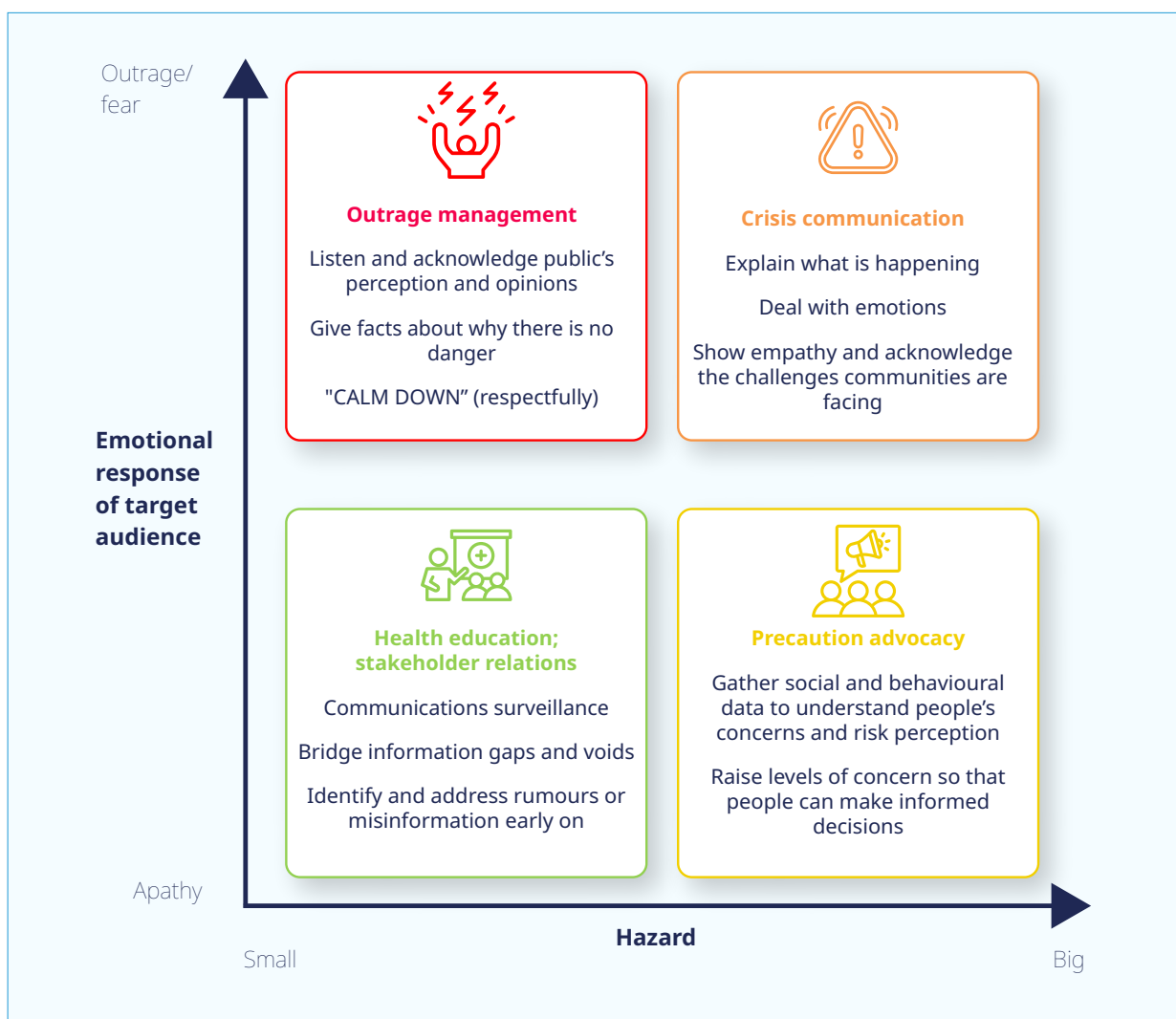
Social listening tools can support the detection of early signs of outrage and help anticipate and manage misinformation cycles.

The fourth strategy is **crisis communication** when the hazard is large or imminent, and fear is also (appropriately) high. The aim of crisis communication is to ensure that transparent, accurate and current information reaches the audience while managing rumours and misinformation and encouraging behaviour change among the population. This method offers an opportunity to promote transparency and share information promptly. Telling people what you know (and what you don't know), what is being done, when more information will be shared, and providing guidance on how to act or

what to do are essential for building trust and reducing outrage. This strategy often requires resources such as social mobilization, social media, mass media, trustworthy spokespersons, etc.

Fig. 7 gives an overview of the key actions and goals for each of the four strategies. It is important for RCCE practitioners to note that the strategy used can, and likely will, change over the course of the readiness and response phases. Communication should be dynamic and reflect the risk assessment and the perception of risk by the communities and various audiences, adapting as needed.

Fig. 6. An example of outrage management in the context of COVID-19 vaccine safety

Fig. 7. Overview of the key actions and goals for each of the four strategies

Tool 7: RCCE readiness and response checklist for incidents involving substandard and falsified medicines

This tool is designed to assist RCCE professionals and responders to update or develop readiness and response plans for high-risk incidents involving substandard and falsified medicines. It provides a comprehensive list of activities that should be considered during both the readiness and response phases related to incidents involving substandard and falsified medicines. Links to additional tools are found in Section 3. If action planning and implementation begin during the response phase, items listed under readiness should also be referred to.

This checklist is adapted from the following documents:

- [10 steps to community readiness](#) (3)
- [Joint External Evaluation tool IHR \(2005\) – third edition](#) (4)
- [International Health Regulations \(2005\) – third edition](#) (5)
- [COVID-19 global risk communication and community engagement strategy: interim guidance](#) (30)
- [Risk communication and community engagement readiness and initial response for novel coronaviruses \(nCoV\): interim guidance](#) (31)
- [Strengthening WHO preparedness for and response to health emergencies: strengthening the global architecture for health emergency, preparedness, response and resilience](#) (32)

Table 7. RCCE readiness and response checklist

Area of work	Steps	Activities
Systems and coordination	Readiness	☐ Establish or strengthen RCCE coordination mechanisms, including establishing an inter-agency task force or crisis communication centre, technical working groups for key areas of work, and ensure content clearance and information sharing protocols are approved. ☐ Review and update existing RCCE strategies and plans using intelligence from local surveillance, epidemiological and social-behavioural data (see Tools 3 and 4). Ensure these are linked to broader emergency preparedness and response plans. ☐ Establish data-sharing agreements with regulatory and customs authorities to facilitate timely RCCE response and promote cross-pillar information flow in substandard and falsified medicine cases. ☐ Set up or strengthen an RCCE team, define members' roles and responsibilities and how the team will link to other response pillars. ☐ Map RCCE expertise at all levels, with specific focal points within the NMRA, ministries of health, law enforcement and local health authorities, critical for multilingual contexts with varied literacy levels. ☐ Conduct or update the PESTEL situational analysis and stakeholder analysis and risk analysis. ☐ Develop a budget, with funding options and a human resource plan, including plans for surge support if needed. ☐ Map and review potential parallel investments to optimize the reach and effectiveness of RCCE activities, especially those that may increase access to licensed and quality-assured medicines. ☐ Document and review decision-making timelines for regulatory actions during the incident to improve accountability and post-incident learning.
	Response	☐ Convene and coordinate the RCCE response with government, stakeholders, partners and across technical areas/pillars. ☐ Liaise with NMRA to provide the most updated information and next steps regarding the status of the medicine, including potential registration status (suspension, cancellation, etc.), recalls, substitute medicines, and any necessary public notice. ☐ Activate the inter-agency task force or crisis communication centre and ensure content clearance and information sharing protocols are followed. ☐ Revise and update RCCE strategies and plan according to need and current surveillance, risk assessment (see Tool 5) and social-behavioural data (see Tool 3), new evidence or learnings and community insights. ☐ Implement approved operational budget and human resource plan, including deployment of surge staff.
Community data for action	Readiness	☐ Conduct a review of social-behavioural data (see Tools 1 and 3) and identify vulnerable populations (see Tool 1), risk factors, priority behaviours and potential barriers and enablers for an effective response (see Tool 4). Use this knowledge to inform decision-making at all levels. ☐ Ensure mechanisms for community listening are established (both online and offline) and respond to rumours and misinformation proactively (see Tool 9 to support tracking of rumours and misinformation). ☐ Analyse gaps in available social data. A mix of quantitative and qualitative data is best, including community feedback, social listening, polling, situational and behavioural analyses, and survey data to understand community knowledge gaps, perceptions and behaviours. Commission appropriate research to fill in the identified gaps. ☐ Set up a framework for MEL to track the efficacy of RCCE activities and impact made. Use findings to tailor and adjust the RCCE strategy and plans accordingly.

Area of work	Steps	Activities
Community data for action	Response	☐ Continuously conduct data collection among at-risk and affected populations to track changes in knowledge, attitudes, perceptions, behaviours and other social-behavioural variables. ☐ Regularly conduct community listening (see Tool 9). Use the findings to develop, adjust and implement RCCE interventions that address concerns, misconceptions, rumours and barriers to uptake of protective behaviours. Address any unacceptable behaviours, including sexual misconduct. Include affected communities throughout this process. ☐ Continue to monitor the impact of response activities on communities (see Tool 9). Ensure plans are in place to manage potential or unexpected impacts (changes to health-seeking behaviours, impact on job and food security, other economic or social impacts) and update accordingly. ☐ Share data back to communities and update local response activities as new social, behavioural and anthropological data become available.
Risk communication	Readiness	☐ Ensure that the highest levels of government are ready to release information to protect the public's health in a rapid, transparent and accessible manner. ☐ Create or review a repository of existing RCCE materials such as message banks, tools, products and templates. ☐ Map and prioritize trusted and commonly used communication channels and platforms. Assess these for accessibility to people in remote areas, without digital skills or access, those with low literacy skills or who may not speak the dominant language, etc. ☐ Identify alternative communication channels to reach all pockets of society, such as street radio, mobile announcers, voice messages for health centres, etc., and partners who can potentially support the dissemination of key messages through these methods. This could include health worker professional associations as trusted intermediaries for both public and private sector practitioners. ☐ Identify focal points and media spokespeople for all key partners at all levels; list their areas of expertise in relation to the disease or health emergency threat; if necessary, train them. Training should consider how to handle hostile or biased/misinformation questions and prepare spokespeople for high-stress situations. ☐ Coordinate communication activities and use standard operating procedures for clearance and sharing. ☐ Ensure that a crisis communication template is developed and that there are clear protocols regarding a high-risk incident involving substandard and falsified medicines.
	Response	☐ In collaboration with affected communities, continuously develop, adapt and test messages based on the perception of risk and as the situation evolves. ☐ Update interventions and messaging, based on the MEL framework and feedback from communities. ☐ Continue to build and deliver high-quality information to raise knowledge and awareness and manage risk perceptions related to the specific topic of interest, using trusted and commonly used channels. ☐ Prepare fact-based materials for anticipated misinformation narratives to speed up rumour countering in high misinformation contexts. ☐ Engage regularly with and provide risk communication content to government, media and other partners to ensure public information is adapted and consistent with the latest science and current context. ☐ Activate spokespeople and influential individuals, including those from other agencies and stakeholders, to align messaging and to broaden the reach of RCCE activities. ☐ Provide guidance to media outlets on how to access reliable information and manage misconceptions.

Area of work	Steps	Activities
Community engagement	Readiness	☐ Hold discussions with communities to understand sociocultural contexts and power dynamics of key audiences. ☐ Identify what type of engagement is safe, feasible and acceptable for different communities. ☐ Identify existing platforms (community leaders, CSOs and key influencers, particularly those accessed by people at risk) and engage communities in decision-making processes. ☐ Establish or strengthen community feedback systems to ensure community beliefs, questions, concerns and suggestions are heard. ☐ Co-develop priority actions with affected groups to raise awareness and knowledge on contaminated medicines and build trust and encourage uptake of protective behaviours. ☐ Design and co-implement interventions and strategies with communities. ☐ Train community engagement teams, including volunteers, and establish surge capacity mechanisms, including on adverse event reporting education to build public participation in surveillance. ☐ Ensure translation capacities are available to tailor all RCCE materials into local languages and dialects. ☐ Anticipate special information and engagement needs for people who are disabled, illiterate or marginalized.
	Response	☐ Update and co-implement RCCE interventions and strategies with communities. ☐ Ensure continuity of community feedback systems and close information gaps. ☐ Launch or strengthen an “alliance” of influencers and stakeholders who can listen, advocate, inform, address rumours and misinformation and promote health literacy using evidence and data. ☐ Ensure representation of civil society and vulnerable groups. Work closely with other committees and advisory groups. ☐ Engage relevant sectors (government, social and private sector) to manage service and supply needs, assess barriers and strengthen referral systems. Ensure affected communities are linked to referral systems.
Capacity building	Readiness	☐ Conduct a rapid needs assessment, which includes mapping of existing RCCE human resource capacities and capabilities. ☐ Develop a capacity plan with stakeholders based on the result of the needs assessment. ☐ Build the capacity of RCCE teams and other key stakeholders based on the plan developed. ☐ Hold simulation exercises involving substandard and falsified medicines scenarios to test coordination and communication readiness. ☐ Create standard operating procedures to drive consistency and quality across RCCE interventions and collaboration with partners. ☐ Initiate a continuous peer-to-peer support system for community mobilizers, responders and networks.
	Response	☐ Adapt capacity-building tools as needed. ☐ Identify and train emergency RCCE staff and potential surge staff on plans and procedures. ☐ Provide refresher or on-the-job training for RCCE responders and spokespersons as interventions and strategies change. ☐ Continue to provide orientation to media professionals and communication networks as the response evolves.

Area of work	Steps	Activities
Measurement, evaluation, and learning (MEL)	Readiness	☐ Develop/review the MEL framework, including monitoring and evaluation indicators based on the developed RCCE strategy, planned activities and expected outcomes (see Tool 11). ☐ Develop/strengthen a real-time monitoring system using existing/adapted tools such as mobile and manual data collection methods, interactive dashboards and automated data analysis. ☐ Train the RCCE team on the use of relevant tools. ☐ Promote community participation in developing the MEL process. ☐ Develop a system to effectively store, manage and share information and key datasets.
	Response	☐ Continuously revise the MEL framework to ensure it is capturing the data needed to measure results and impact (see Tool 11). ☐ Use established real-time and participatory monitoring and evaluation systems where possible, such as mobile or application-based reporting. ☐ Generate evidence and data that allows regular assessment of strategy implementation and impact. ☐ Capture qualitative feedback from community leaders and pharmacists in addition to quantitative data to support contextual insights. ☐ Include NMRA and other key stakeholders in monitoring, reporting and joint accountability efforts to increase the likelihood of broad community uptake and responsibility for new interventions. ☐ Maintain and strengthen systems to effectively manage and share information, document lessons learned and gather best practices. Disseminate lessons and best practices widely.

Tool 8: Activities tracker

This tool is designed to assist RCCE decision-makers, practitioners and partners to track activities once identified using the RCCE readiness and response checklist (Tool 7). Table 8 shows an example activity.

Table 8. Activities tracker

Area of work	Task/activity	Priority level	Organization and individual responsible	Budget/resources	Links	Deadline	Status
Community engagement		High, medium or low	Ministry of health; name, email, phone number		To any working documents or webpage		Complete/ in progress/ incomplete

Implementation

The tools in this section are designed to support activities conducted as part of evidence-based RCCE strategies and plans. While the projects and activities that need to be implemented will vary in each context based on needs and strategy, these resources offer ways to approach some key components of most RCCE plans. Communities should be considered key implementing partners for RCCE activities during incidents involving contaminated medicines.

Community listening encompasses various approaches to collecting data to identify current narratives, questions, rumours, misinformation, levels of trust and other relevant factors from at-risk populations. It can help to track and monitor trends, changing attitudes towards health authorities and interventions, and identify newly emerging concerns.

Tool 9: Community listening and feedback systems for incidents involving substandard and falsified medicines

On- and offline sources should be used for community listening. Offline sources of data can include community feedback systems, qualitative interviews, focus group discussions, pharmacovigilance reporting forms, hotline data, findings from social-behavioural research, and television and radio. Online sources can include social media, websites, chat rooms, etc. All community listening sources have advantages, biases and limitations, which should be documented when reporting data.

To collect community feedback about substandard and falsified medicines, identify community representatives who are closely involved with readiness and response activities or who are from or represent affected communities. CSOs already involved in related health advocacy or service provision (e.g. those developing and distributing information, education and communication materials) are good sources of community feedback, as these groups can provide targeted input and help reach specific demographics more effectively. In support of collecting relevant feedback and conducting structured and effective conversations with communities, WHO has developed the [Community conversation kit](#) (33) to help people who have a leadership role speak with people in their community about how to protect themselves from a health threat.

To effectively use community listening in managing an incident involving substandard and falsified medicines, health authorities and all involved, partners should use the full range of on- and offline tools to collect, monitor and analyse public narrative and conversations related to the contaminated product. These tools may vary significantly from context to context and regarding specific community needs, access and norms. Particular attention should be given to key populations and themes of misinformation, while also identifying information voids such as the sudden increase in searches for e.g. "[name of medicine]", "contamination" or "[name of medicine] counterfeit/falsified/fake".

Setting up a dedicated social listening system involves defining objectives, selecting relevant social media platforms, identifying substandard and falsified medicines-related keywords and hashtags, setting up a taxonomy, monitoring these keywords using tools like WHO Epidemic Intelligence from Open Sources, tools that analyse the popularity of search queries across various regions and languages, conducting data analysis, and reporting the findings to stakeholders. The system should be regularly reviewed and adjusted based on the findings, such as adding new keywords, hashtags or identifying new platforms of concern. Table 9 is a template to support analysis of the collected data from a variety of sources.

Table 9. Template to support analysis of collected data

How is the population:	Finding	Source and date	Context	Action needed
Feeling	Parents are fearful and distrustful of paediatric paracetamol medicines after child deaths linked to syrup medicines	Al Jazeera – <i>Gambia parents 'fight for children' in landmark trial on India syrup deaths</i> , November 2023	Over 60 children died in The Gambia due to toxic contaminants (diethylene glycol) in imported syrups	Rebuild trust through transparent communication, strengthen pharmacovigilance and ensure safe alternatives are available
Thinking	People believe they are buying genuine weight-loss medication online, but are receiving falsified products	WHO Medical Product Alert No.5/2023	Falsified Ozempic pens containing insulin instead of semaglutide were found in Europe	Raise awareness about risks of buying medicines online; promote use of verified pharmacies
Behaving	People are increasingly purchasing medicines from unverified online pharmacies	WHO and national regulatory authorities, 2023–2024	Driven by convenience, cost and anonymity, many consumers unknowingly buy falsified or substandard medicines online	Promote awareness campaigns on safe online purchasing, encourage use of verified pharmacy platforms, and strengthen digital surveillance

The data sourced through both social listening and community feedback systems can be triangulated with epidemiological data, research and programme data to gain additional insights to inform strategy and planning.

The development of community listening and community feedback systems for readiness and response will depend on existing platforms and resources but should be considered an essential part of any RCCE planning. The following resources can be used to inform activities, as can the matrix in Table 10.

- [Community engagement in humanitarian action toolkit \(CHAT\) \(34\)](#)
- [IFRC feedback kit \(35\)](#)
- [How to build an infodemic insights report in six steps \(36\)](#)
- [Infodemic management training 101 \(OpenWHO\) \(37\)](#)
- [Infodemic management: defining a taxonomy for social listening \(OpenWHO\) \(38\)](#).

Table 10. Rumours, misinformation and event tracker

Issue/ event/date	Country of origin	Platform (print, web, social media, official statement, etc.)	Level of risk (low or high)	Facts (what really happened, scientific explana- tions, etc.)	Respond YES/NO	Initial response (if YES) of (insert name of agency or organiza- tion) (key messages)	Cleared by:
							Date:
							Date:
							Date:
							Date:
							Date:

Tool 10: Developing key messages and content

This tool is designed to guide the development of key messages based on data collected, social-behavioural insights, epidemiological surveillance and best practices. Key messages are the main points of information you want to convey to the audience so that they will understand and remember the risk of taking substandard and falsified medicines.

These should be clear and concise statements that explain key concepts and factual information in lay language. Key messages should also support the desired communication outcomes – the change needed in the behaviour of the affected population.

Since a key component of prevention is education and awareness, patients and consumers need to be provided with sufficient information to be able to make an informed decision on how to avoid substandard and falsified medicines and what to do if they believe they are exposed to one. When substandard and falsified medicines are detected in the supply chain, a timely, proportionate and effective regulatory response is required, as is a communications response. From health care professionals and supply chain professionals to law enforcement, e.g. police, customs and the judiciary, all need to be made aware of the existence, risks and impact of substandard and falsified medicines including:

- the dangers of substandard and falsified medicines;
- where they are found and how they can reach consumers;
- protective behaviours that can be taken;

- action being taken by the relevant authorities;
- how to identify substandard and falsified medicines;
- how to report a product; and
- what to do if you get ill.

When developing messages for the general public, it is important to cater to both the mind, by providing timely, accurate and transparent information, and heart, doing so respectfully and with empathy and care.

Gathering as much information as possible about the setting in which the emergency is occurring, the individual or community perception of risk, and the ability to act are essential to appropriately arouse or subdue emotions to manage outrage and drive action while maintaining trust. Key messages need to be adapted based on what is known about key audiences (Tool 1); the local context (Tool 3); enablers and barriers of key behaviour (Tool 4); risk assessment (Tool 5); and what is being learned through community listening (Tool 9).

These tools all contribute to the collection of social-behavioural insights and other data to determine the key influences (cognitive, social

and environmental) on the audience and the specific behaviour to target. This sets the foundation for developing clear and actionable messages, and with the application of a few more communications principles you will be able to develop messages that help the audience to understand risks, enhance their protection against an incident involving substandard and falsified medicines, improve health literacy, close information gaps and combat mis- or disinformation in a timely and adaptable manner.

Key steps for developing messages include:

- developing a single overarching communications outcome (SOCO);
- framing messages to support uptake;
- creating a message map;
- application of the “7 Cs”;
- deciding on appropriate channels for message delivery; and
- pretesting messages to ensure they are as effective as possible.

Developing a single overarching communications outcome

The SOCO should be the change you want to see in the audience as a result of effective messaging. It considers the specific target group and their risk perception, the issue/behaviour that needs to be addressed, and why it is important to address this behaviour or issue at this point in time. It provides greater clarity about the communication priority and aids in keeping messages focused and structured (39). It is important to remember that a SOCO is an outcome, not an objective, that the SOCO should be expressed from the perspective of the audience, and that it is time-limited, realistic and achievable.

Example of a SOCO: People are practising safe purchase of medicines through verified sources.

Framing messages

All messages should communicate risks, contain public health advice, and be action-oriented to drive behaviour change and reach the desired outcomes. The way messages are framed determines how the information is presented and what is emphasized, and often looks at how to best structure and communicate health risks and recommended action (40, 41). The approach used should be informed by the social-behavioural and other data collected. Three types of framing that are particularly effective at motivating an audience to change behaviours are:

- **self-efficacy:** an individual's belief in their ability to succeed in attaining a certain goal or outcome;

- **pro-social norms:** clear, healthy, ethical standards, beliefs and behaviour guidelines that promote pro-social behaviour and minimize health risks; and
- **gain or benefit:** highlighting the advantages associated with practising the desired behaviour.

When using **self-efficacy framing**, people with a strong belief in their ability to complete the behaviour are more likely to comply. If the desired behaviour is seen as difficult or personal action is considered inconsequential, there is very little motivation for them to engage in it (42).

Self-efficacy framing example: “You can reduce your risk of getting dengue virus with simple actions: keep your body covered, reduce possible mosquito breeding sites, and limit your time outdoors when mosquitoes are most active.”

The behaviour of individuals living in communities is also very much regulated by religious, social and moral norms and values – ultimately, people tend to align their behaviour with what they think others are doing or what they think others approve or disapprove of. This pro-social framing aligns with the individual's moral values, appeals to social consensus or norms and highlights social benefits and group approval (43, 44).

Pro-social framing example: “Millions of people have been vaccinated against COVID-19, will you join them? Get your vaccine or booster shot today.”

Lastly, various global research also demonstrates that often health messages framed around gain or benefit increase intention to adopt self-care behaviours and have the strongest impact, whereas loss-framed health messages can increase risk perception (45–47).

Gain or benefit framing example: “If you quit smoking, you will experience immediate and long-term health benefits. Kick the habit and improve your health!”

Message mapping

A message map is a tool to align messaging among different audiences, prioritize and structure messages clearly and consistently, and can help identify and close information gaps. The standard message map structure involves stating a claim and targeting a specific behaviour with a key message, then developing three supporting messages to bolster the key message and reinforce the initial claim, as seen in Table 11.

Table 11. Key messages and supporting messages template for high-risk incidents involving substandard and falsified medicines

Key message	Protect your health and the health of your community – only purchase and use medicines from verified sources.
Target behaviour	Avoid purchasing medicines from informal markets or unverified online sources.
Supporting message 1	You can protect yourself and your family by using medicines that have official approval and buying only from licensed pharmacies.
Supporting message 2	If you use verified medicines, you reduce your risk of treatment failure, poisoning and long-term health complications.
Supporting message 3	Millions of people are choosing safe, approved medicines – join them and help keep your community safe.

To increase the likelihood that the target audience is receptive and pays attention to the messages, you can apply the “27/9/3” rule:

- The average length of a sound bite in print media is 27 words.
- The average duration of a sound bite in broadcast media is 9 seconds.
- The average number of messages reported in both print and broadcast media is three.

Apply the 7 Cs of effective communication

This set of principles is core to the development of effective and well-constructed messages (48). It is a simple list of important characteristics that the messages should have, and is essential in situations where information is not necessarily positive, but is still critical to share:

- **Clear:** be straightforward about the goal or message, avoid jargon.
- **Concise:** stick to the point and keep it brief and precise.
- **Concrete:** build a strong message with a few key facts or details, be specific.
- **Correct:** communicate at the level of the audience.
- **Coherent:** the message is understandable, related to the main topic and consistent.
- **Complete:** the provided information enables the audience to take action and make informed decisions.
- **Considerate:** communication is open and honest, be empathetic.

Message delivery

The selection of channel(s) for the delivery of messages and high-quality information to the specific audience is extremely important. It must be appropriate for local audiences, based on a strong understanding of:

- How the audience communicates.
- Various social and cultural characteristics which define how the audience receives information, such as language spoken, levels of literacy and education, and more.
- Access to and use of different platforms or channels.
- If the channel is a trusted source of information.

Appropriate channels

Channels can include and are not limited to:

- **Traditional media**, such as television, radio, newspapers and billboards.
- **Social media** including Facebook, X, Instagram, Google+, blogs and more.
- **Online platforms** and **websites** of trusted organizations/sources and partners.
- **Community resources** such as community health boards or local community centres, influencers and community leaders, including health care workers.
- **Mobile phones** for SMS services, chatbots and messaging applications like WhatsApp, Viber and WeChat (49).

Pretesting messages

At this stage, you know what you want to communicate – the target behaviour you are aiming for. You have developed messages highlighting who needs to do what, when, where and how, and identified the platforms to be used to reach the target audience. Next comes testing the messages to see if they will be successful.

This involves bringing together members of the target audience to review and provide feedback on draft messages and other elements of communication in an effort to increase implementation and uptake of the final messages. This can be done through focus group discussions with community members or by conducting a large-scale survey. Pretesting can provide useful insight into how a behaviour is understood or a message received. This feedback is extremely important as it allows RCCE practitioners to co-design the solutions and final products with the affected communities, to ensure maximum

comprehension and uptake of the protective behaviours (50). This includes identification of appropriate languages and expressions to ensure messages are culturally relevant and meaningful, and the involvement of community leaders or local health workers to guarantee relevance and support trust and acceptance within the audience.

Table 12 is a checklist with key considerations for developing effective messages, tailored to the specific needs of the target audience. Additional information on how to test messages can be found in WHO's *Message testing during an epidemic or pandemic* (51).

Table 12. Checklist for developing effective messages

Step 1: Know the audience

- ☐ Identify the audience(s) that are being communicated with (people at risk, health care workers, community volunteers, etc.).
- ☐ Consolidate all available sources of information to form a clear picture of the audience(s), including level of knowledge, health literacy, access to information, etc.
- ☐ Tailor messages based on identified information gaps, local needs or specific requirements of the audience.
- ☐ Make sure to be aware of any sensitivities related to the substandard and falsified medicine incident or the community itself.

Step 2: Identify and target a specific behaviour

- ☐ Define a single behaviour that needs to be changed to achieve a desired public health outcome (handwashing, vaccination, wearing a mask, etc.).
- ☐ Consider the audience's ability and competence to perform the behaviour.
- ☐ What knowledge or understanding gaps might limit someone from performing the desired behaviour?
- ☐ Are there any barriers that could impact the uptake of the behaviour?
- ☐ Think about whether the behaviour is realistic and how it may be impacted by social norms, environmental context and overall resources.
- ☐ Do people have access to health services or other resources required to perform the behaviour?
- ☐ Reflect on possible influences on the audience's motivation – what are their expectations or perceived consequences? What's in it for them and what are the benefits of performing the behaviour?
- ☐ Is the behaviour associated with positive or negative feelings? How does this impact the uptake of the behaviour?

Step 3: Define risks and communicate clear public health advice

- ☐ Based on the risk perception of the audience, set the tone of the message to arouse or subdue outrage for maximum uptake of protective behaviours.
- ☐ Clearly communicate what the health risks are and actions that can be taken to mitigate these threats.
- ☐ Provide a brief explanation or reason why a behaviour should be performed.
- ☐ Define clear steps the audience should take to protect their health.
- ☐ Include a clear action that directly conveys what people should or should not do.
- ☐ Make sure public health advice is actionable and in line with WHO recommendations and guidance.

Step 4: Develop the single overarching communications outcome (SOCO)

- ☐ Identify the health problem that needs to be solved.
- ☐ Identify what needs to change now to make progress on this health problem possible.
- ☐ Specify who needs to take action to make this change possible.
- ☐ Define what the audience needs to change or do as a result of communication.

Step 5: Framing messages for maximum comprehension

- ☐ Depending on the topic, local context and available information about the audience, use one of the three recommended framing models – self-efficacy, pro-social or gain/benefit.
- ☐ Use self-efficacy framing with groups who already have established competence or an adequate level of health literacy to perform the behaviour, e.g. health workers or doctors.
- ☐ If using pro-social framing, ensure it is reflective of and aligned with the community or audience's religious, social or cultural values.
- ☐ Use pro-social framing to encourage people to undertake a behaviour for the benefit of others.
- ☐ Use gain or benefit framing to communicate the added value of the behaviour and positive health outcomes.

Step 6: Create a message map

- ☐ Design a key message that communicates risk, public health advice and a clear call to action.
- ☐ Think about what additional information you can provide in the supporting messages to close information gaps or address rumours and misinformation.
- ☐ Apply the 27/9/3 rule as often as possible for more concise messages.
- ☐ Make sure the targeted behaviour is consistent throughout the message map.

Step 7: Apply the 7 Cs

- ☐ Be straightforward, communicate what is known and what is not known, and avoid overly technical language and jargon.
- ☐ Break down numbers using “social maths” – translate data and statistics or communicate complex, number-based messages by applying them to something meaningful, e.g. 303 000 women die each year from pregnancy and childbirth complications. That’s 34 women dying per hour.
- ☐ Use as few words as possible (aim for a maximum of 27 words), while still conveying the importance of the matter.
- ☐ Share key facts and details, be specific.
- ☐ Break down messages to be consistent with the level of health literacy, knowledge or perception of risk of the audience. There is no such thing as common knowledge – do not assume the audience is already familiar with the topic.
- ☐ Be consistent across all messages. Ensure data, facts and figures are accurate and align across the message map.
- ☐ Provide enough information that the audience can make informed decisions and can take action, and change/adapt behaviour.
- ☐ Cater to the minds and hearts of the audience with open, honest and empathetic communication.
- ☐ Translate materials into multiple languages when needed, and be sure to use culturally sensitive and appropriate language.

Step 8: Message delivery

- ☐ Based on community listening and PESTEL analysis findings, deliver messages through the channels most commonly used by the affected communities.
- ☐ Ensure that the messages are translated into the local languages to reach all groups of the population affected by the crisis.
- ☐ Involve community leaders and key influencers to amplify the messages.
- ☐ Create different formats of messages (audio, video, print) to reach all pockets of society.
- ☐ Ensure that messages are in alignment with the requirements of the platform you are using (length, style, format).
- ☐ Consider adding a visual sample (e.g. banner or footer) indicating official regulatory notice with date/time stamp and QR code to official web page, to help the public distinguish official vs fake notices.

Step 9: Pretesting messages

- ☐ If possible, conduct quantitative testing of messages to identify the best performing ones before mass rollout.
- ☐ If time is limited, undertake rapid qualitative testing to optimize message content and presentation and minimize the risk of backfire or unforeseen challenges with messages.
- ☐ Ensure diversity within the testing audience to best represent the affected or vulnerable communities.
- ☐ Integrate key findings and refine the messages based on feedback.

Tool 11: Measurement, evaluation and learning (MEL)

This tool will help enhance the accountability and effectiveness of RCCE through measuring, evaluating activities and constantly learning from audiences how to improve or adapt interventions to achieve expected health outcomes.

A MEL framework recognizes the importance of:

- measurement to collect evidence;
- evaluation and systematic analysis of results; and
- learning to gain insights and new knowledge that can be applied in future planning and strategy.

MEL should be used throughout all phases of the emergency management cycle and should include community participation to support sustainability, joint accountability and ultimately increase the effectiveness of RCCE strategies, plans and interventions (52). Active community participation should not be limited to the initial planning phase but must continue throughout implementation and evaluation to ensure interventions remain relevant and culturally appropriate.

Once you have determined if the problem you are tackling is behavioural in nature or if it is another type of barrier, such as environmental or structural, it is possible to design interventions. Many different models can help design and structure the MEL framework, based on priorities or targeted behaviours. Within the *MEL manual*, WHO proposes the “theory of change” and “program logic models.” For more detailed information on these tools and others, see [The MEL Manual](#) (52).

The theory of change and program logic models help logically explain how the intervention is expected to lead to the desired behaviour change and how to measure it along the way. The theory of change involves two key steps:

- **Step 1:** Identification of all the possible interventions and/or stimuli that can lead to a change in a particular context.
- **Step 2:** Examination of the evidence and assumptions that support such beliefs.

The program logic model helps demonstrate the theory of change by linking activities with outputs, and short-term and longer term outcomes (see Table 13). The next step is to develop specific, measurable, achievable, realistic and time-bound (SMART) objectives and indicators to measure the progress and impact of the intervention. Qualitative and quantitative indicators should be identified and collected at each stage of RCCE activities and aligned with national plans to reflect priority actions and desired outcomes. Table 13 provides a helpful template for structuring and planning a MEL framework and to inform the identification of indicators that are fit for the local context.

The Collective Service developed [Risk communication and community engagement indicator guidance for COVID-19](#) (53), which provides useful support that can be applied to other emergencies, including incidents involving substandard and falsified medicines.

Table 13. Measurement, evaluation and learning framework template

Theory of change (example)						
People are unaware of the risk posed when buying medicines through unverified online markets.	People are informed about the risks associated with buying medicines through unverified online markets.	People form an opinion about unverified online markets and potential access to substandard and falsified medicines.	People are consistently engaged in on-line and offline conversations about the risk of purchasing medicines through unverified online markets.	People acknowledge the value of purchasing medicines through verified sources, online or otherwise.	People support measures to avoid substandard and falsified medicines by purchasing through verified sources.	People are practising safe purchase of medicines through verified sources.
What do you need to complete MEL (example)			When and how should you report on findings			
1. Situational analysis (PESTEL)			Here you should briefly outline a reporting plan, including reporting intervals, format, general content and more.			
2. Risk assessment						
3. Behavioural analysis						
4. Community listening						
5. Stakeholder analysis						
6. Community feedback mechanisms						
7. Social listening reports						
8. Access to Google analytics or other tools related to social media						
What will you track?						
Inputs		Metrics and indicators		Methods		
Data and insights collected through various proposed research can qualify as inputs						
Activities		Metrics and indicators		Methods		
Producing and distributing RCCE products based on the collected data and insights						
Outputs		Metrics and indicators		Methods		
Reaching and engaging audiences						
		# of information, education and communications (IEC) materials developed		Log of materials in circulation (quantitative)		
		# of posts on social media		Content analysis and social media reports (qualitative)		
		# of announcements released		Log of radio and TV announcements (quantitative)		
Short-term outcomes		Metrics and indicators		Methods		
Assessing audiences' initial reactions, response to RCCE activities						
Long-term outcomes		Metrics and indicators		Methods		
Evaluating what sustainable effects RCCE activities had on audiences						
Impact		Metrics and indicators		Methods		
Evaluating the results achieved, in full or in part, by RCCE activities						

4. Principles and considerations for high-risk incidents involving substandard and falsified medicines

This section contains additional considerations for RCCE strategy, planning and implementation during high-risk incidents involving substandard and falsified medicines. Table 14 provides a timeline for RCCE activities. More on RCCE principles can be found in the [10 steps to community readiness](#) (3).

Addressing uncertainty and maintaining trust

Managing uncertainty is an important function of RCCE during health emergencies. Poor management of an incident leads to difficult-to-explain delays and the potential for increased harm to patients. Unclear or insufficient communication can lead to a more serious health event than the initial incident that was reported. Both lead to a loss of confidence in medicines, health systems, health care professionals, and organizational reputational damage. Therefore, it is essential that strategies and plans for RCCE incorporate approaches for managing and addressing uncertainty to maintain public trust throughout the emergency. It is important to ensure communications are aligned with the perception of risk and are actionable – that the public is warned and prepared in a manner that avoids increasing panic and drives action. Key steps for managing uncertainty include:

- assessing the situation to understand what is known and unknown;
- identifying key uncertainties that may impact community understanding and response;
- being transparent and honest;
- setting realistic expectations;

- providing timely updates when new evidence emerges;
- explaining what is being done to find out more;
- acknowledging that what is known may change;
- listening and responding to community concerns; and
- being prepared to adapt (54).

RCCE activities should also focus on:

- **Equitability:** to ensure the consideration and inclusion of the needs of those whose vulnerabilities to substandard and falsified medicines are higher.
- **Parallel investments:** to optimize the effectiveness and equitability of activities within the target population.
- **Sustainability:** particularly when activities may need to be continued over a long period of time to reach the population most effectively.

Table 14. Timeline for RCCE activities following an incident involving substandard and falsified medicines

Timing	Action
Before incident	• Conduct situational analysis (PESTEL), behavioural research, community listening and stakeholder analysis. • Develop “rumours, misinformation and event tracker”. • Assemble background information about the substandard and falsified medicines subject to the public health situation. • Prepare materials (Q&As, facts sheets, talking points, etc.). • Build relationships with media, partners and local health authorities. • Provide ongoing information and raise awareness with the media about the subject of substandard and falsified medicines. • Train relevant staff and spokespersons.
	Immediately:
	• Identify what has happened and verify the report in close coordination with experts within the NMRA and other key stakeholders. • Gather information and analyse data. • Alert other pillar leads from the IMST and relevant partner focal points. • Decide the level of risk and whether to communicate forward. • Use community and social listening finding to inform decision-making
	Within 24 hours:
	• Prepare a response with inputs from technicals and other subject matter experts. • Prepare talking points and background data. • Coordinate response with the ministry of health, local health authorities and partners as relevant. • Select the potential communication channel based on local context (radio, TV, print, etc.).
	Within 72 hours:
	• Consider a press release. • Consider a joint press conference (NMRA, ministry of health, WHO and other involved partners).
During incident	Ongoing:
	• Provide information to health care workers, partners, media and the public. • Update with interim information until definitive results are available.
	After incident
After incident	• Evaluate communication approach and efficacy of applied protocols, adjust as needed. • Provide ongoing information to the media and the public about the specific substandard and falsified medicine and substandard and falsified medicines in general.

5. Key tools, products and training resources for incidents involving substandard and falsified medicines

Guidance	
<i>WHO prevent–detect–respond strategy to combat substandard and falsified medical products</i> (55)	An overview of prevention, detection and response measures to substandard and falsified medical products, including overall objective, actions and impact at each phase of the emergency management cycle. This includes considerations regarding education and awareness, stakeholder engagement, reporting systems, surveillance, regulatory strengthening, legal procedures and more.
<i>The WHO Member State Mechanism on Substandard and Falsified Medical Products, update 2022</i> (10)	Substandard and falsified medicines are produced, distributed and sold all over the world. Preventing, detecting and responding to them is a persistent public health challenge for WHO Member States. This document outlines how WHO Member States work together to increase access to safe, effective and quality medicines, vaccines and other medical products.
<i>WHO Member State Mechanism</i> (11)	Background information and resources regarding the Member State Mechanism, which aims to order to protect public health and promote access to affordable, safe, efficacious and quality medical products, and through effective collaboration among the Member States and the Secretariat, promote the prevention and control of substandard and falsified medical products and associated activities.
<i>WHO Global Surveillance and Monitoring System for substandard and falsified medical products</i> (13)	This report, based on data gathered by the WHO GSMS for substandard and falsified medicines, vaccines and in vitro diagnostic tests during its first 4 years of operation, up to 30 June 2017, examines the issue in greater detail.
<i>WHO Global Benchmarking Tools (GBT) for evaluation of national regulatory systems</i> (56)	Regulatory systems play a key role in assuring the quality, safety and efficacy of medical products. The Global Benchmarking Tools (GBT) represents the primary means by which WHO objectively evaluates regulatory systems, as mandated by resolution WHA67.20: Regulatory system strengthening for medical products. The tool and benchmarking methodology enables WHO and regulatory authorities to identify strengths and areas for improvement; facilitate the formulation of institutional development plans to build on strengths and address the identified gaps; prioritize interventions; and monitor progress and achievements.
<i>Recommendations for health authorities on criteria for risk assessment and prioritization of cases of unregistered/unlicensed, substandard and falsified medical products</i> (23)	The document refers to the need for a risk assessment approach in the evaluation of alerts, notifications and reports received by national and regional regulatory authorities (NRRAs). The document was developed with the aim of presenting information on this matter for the consideration of NRRAs, given the mandate of the Member State Mechanism prioritized activity A (drafting recommendations on criteria for risk classification and assessment prioritization of cases of unregistered/unlicensed, substandard and falsified medical products).

Counterfeit medical products & similar crimes: risk communication (57)

A collection of studies and examples in risk communication regarding counterfeit medicines and similar crimes, prepared by experts in the frame of the ongoing programme of activities carried out by the Committee of Experts on Minimising Public Health Risks Posed by Counterfeiting of Medical Products and Similar Crimes (European Directorate for the Quality of Medicines & HealthCare) supported by the Italian Medicines Agency.

Q&As and fact sheets

Regulation and prequalification: regulatory strengthening and capacity building (9)

An overview of WHO's progress towards enhancing regulatory practices related to medical products around the world supported by capacity building, promoting regulatory convergence and harmonization.

Substandard and falsified medical products: key facts (1)

WHO's main fact sheet on substandard and falsified medical products including key facts, scope of the issue, impact and challenges, and WHO response.

Substandard and falsified medical products (14)

WHO main information page on substandard and falsified medical products: overview, impact and links to additional resources, guidance and information.

Contaminated medicines affecting children: questions and answers (15)

WHO information page focusing on the potential danger of contaminated medicines, previous incidents, and guidance, updates and recommendations.

Substandard and falsified medical products: scope, scale and harm (16)

WHO information page detailing the magnitude and possible impact of substandard and falsified medical products.

Public health advice

Substandard and falsified medical products: advice to patients and consumers (58)

WHO information page with consumer guidance and answers to frequently asked questions on the topic of substandard and falsified medical products and additional resources.

Social media and best practices for health authorities (59)

The Committee of Experts on Minimising Public Health Risks Posed by Falsification of Medical Products and Similar Crimes, conscious of the importance of social media, decided to gather information from health regulatory authorities on how best to use social media to promote their own messages and advance their objectives. This document has been drafted to support regulatory authorities at national level regarding their use of social media in communicating with the general public.

Other reference tools

Rapid risk assessment of acute public health events: manual (22)

This manual has been developed to guide rapid risk assessment of acute public health risks from any type of hazard in response to requests from WHO Member States. The manual is aimed primarily at national departments with health protection responsibilities, national focal points for the IHR and WHO staff. It should also be useful to others who join multidisciplinary risk assessment teams, such as clinicians, field epidemiologists, veterinarians, chemists and food safety specialists.

Communicating risk in public health emergencies (49)

This brochure shares examples from the WHO Regional Office for Europe on the crucial role that risk communication played during the COVID-19 pandemic and other emergencies. It argues for investment in risk communication as an integral part of the entire emergency cycle. It highlights building trust and engaging with affected populations, integrating emergency risk communication into health and emergency response systems, and emergency risk communication practice.

A study on the public health and socioeconomic impact of substandard and falsified medical products (60)

The main objective of this study is to provide information on and quantify the cost and socioeconomic impact of falsified and substandard medicines and establish the potential costs and benefits of strengthening regulatory systems to secure the health products supply chain. A second objective is to suggest a method that countries can use to assess the extent of the problem domestically.

"IDEAS": A global communications framework to help combat the threat from substandard and falsified medical products (61)

In order to maintain a global perspective for existing and future communications advice, a communications working group was established and invitations to join the group were extended to communications professionals across all six WHO regions and the broader professional communications community. This group worked to build a contemporary picture of best practice in communications before developing and testing a bespoke approach to guidance on substandard and falsified medical products. The proposals set out in this document are based on the feedback received, together with the results of a global communications survey and live testing of the framework subsequently developed.

Communication for behavioural impact (COMBI) – toolkit (62)

This toolkit will be helpful for people designing more effective emergency response measures. It can be scaled up or down, depending on the situation. It can be applied at subnational and national levels. It was designed for risk communication, developmental communication and health promotion personnel working in multidisciplinary teams to investigate and respond to disease outbreaks.

Training resource

WHO e-course on substandard and falsified medical products (63)

This e-course is designed to equip national health authorities with the necessary knowledge and skills to effectively address the issue of substandard and falsified medical products. Available in English, French and Spanish, this training is part of the range of services offered to focal points through the WHO GSMS.

6. Case studies

Case study 1: Media campaigns on substandard and falsified medical products: a case study in sub-Saharan Africa

Substandard and falsified medical products pose a serious threat to public health around the world. However, their prevalence and effects are unevenly distributed between and within regions and countries (64–66). Some regions are disproportionately affected, typically due to constrained access to available, affordable, quality, safe and effective medicines. This is linked to a lack of proper regulatory and governance systems, inadequate quality control, limited manufacturing capacity and distribution and/or poor supply chain management (1). Within impacted regions, particularly sub-Saharan Africa (64), structurally disadvantaged groups are especially vulnerable to substandard and falsified medical products and their consequences (8, 65, 66). These groups include women, people living in rural areas and individuals with lower socioeconomic or educational status.

Preventing or reducing the risk of exposure to substandard and falsified medical products requires interventions on both the supply side and demand side. On the supply side, strengthening regulatory capacity and ensuring the integrity of supply chains can help to prevent substandard and falsified medical products from reaching consumers. On the demand side, raising awareness and educating communities can empower consumers to protect themselves and others, and make informed decisions against this threat (13).

Public and media campaigns can therefore play a role in informing consumers about steps they can take to reduce their risk of exposure to standard and falsified medical products. From 2020–2021, WHO, in collaboration with local NMRAs, led nationwide pilot media campaigns in Ghana, Nigeria, Sierra Leone and Uganda (8, 66). These campaigns aimed to inform the public about the risks associated with standard and falsified medical products and provide practical guidance on how to identify and avoid purchasing unsafe medicines.

The joint media campaigns were designed using public-facing broadcast media (e.g. TV advertisements and radio jingles or talk shows) to disseminate important information around how the public could minimize their risk of obtaining

standard and falsified medical products. These campaigns aired for a period of 20 days to 3 months, alongside print media (e.g. posters and bumper stickers), distributed throughout each of the four countries. Broadcast media were produced in English and at least one local language; posters were produced exclusively in English.

Although there was variation in the specific contents of each campaign component within and between countries, the public-facing campaign materials broadly advised the public to take the following steps:

- Speak with a health care professional when feeling unwell.
- Only buy medicines from licensed pharmacies or over-the-counter medicine sellers.
- Check medicine packaging for poor condition, spelling mistakes and unclear labels.
- Check medicine packaging for batch number, manufacturing date, expiry date, manufacturer's address.
- Ensure the medicine does not look unusual or broken.
- Report any suspicious medicines, adverse reactions to a medicine, or unlicensed sellers to the local NMRA using the phone and/or WhatsApp number provided on the poster.

Following the pilot, an evaluation was carried out in each of the four countries in 2022–2023 to assess the campaign's overall reach, whether it effectively reached its target audiences (especially those at highest risk) and its impact on behaviour (67). To do this, a series of focus group discussions were conducted with over 600 community members from urban and rural areas across multiple regions in each country, and supplemented by more than 100 key informant interviews in the same locations. In each focus group discussion or interview, participants were shown the campaign materials and were systematically asked if they had encountered them before. This was followed by discussion of the materials, participants' responses and reactions, and what adjustments or strategies might make a campaign more effective going forward.

The post-campaign evaluation estimated the overall reach of the campaign was relatively low. Across the four countries, only 10.8% (range 8.0–13.9%) of participants recalled having encountered the campaign materials. Men were twice as likely as women to have encountered the campaign, and those living in urban areas were 50% more likely than those in rural areas to have seen the campaign.

These findings suggest that broad, nationally targeted media campaigns may fail to effectively reach their intended audience, particularly the most disadvantaged (and most at-risk) groups, without specific efforts to tailor the campaigns to reach and engage them. In particular, mass media communication channels, particularly internet or television, may fail to reach more rural, vulnerable populations who may lack access to these channels.

These same disadvantaged groups were also least likely to be able to understand and interpret the materials appropriately, especially those that required a good level of literacy and/or speaking a majority language. In focus groups and interviews, participants suggested modifications that could help improve a campaign's reach and understandability; for example, delivering campaign messages through trusted community-based channels, such as local or religious leaders, rather than relying solely on mass media. Others emphasized the importance of tailoring the language and format to reflect local dialects and literacy levels to ensure that the right messages reach the right people.

The majority of participants recognized the importance and relevance of campaign messages, but identified significant obstacles to putting them into practice. Each audience has different backgrounds, needs, concerns, enablers and barriers, perceptions, knowledge and experiences that shape both what they understand and how they behave in response to the risk and the information. In these instances, participants noted that inaccessibility and/or unaffordability of quality-assured medicines within their communities were major barriers that limited how they could realistically act on the campaign's messages. Nationally representative survey data from the same four countries corroborate this, indicating that people may engage in "higher risk" practices, such as buying medicines from informal sellers, even when they are aware of substandard and falsified medical products, because they lack realistic alternatives (1).

These findings/feedback reflect deeper structural barriers, such as poor access to affordable, quality-assured medicines, which education alone cannot overcome. It is therefore important to recognize that improving public awareness of substandard and falsified medical products will not necessarily result in behaviour change unless these underlying constraints are addressed at a systems level.

Lessons learned

Identify the behavioural vs structural challenges

Before designing an RCCE campaign, it is critical to determine whether the issue at hand is linked to behavioural factors or stems from broader social, economic or health system challenges. A premature focus on changing individual behaviours without understanding underlying barriers can result in interventions where the target audience is unable to act meaningfully on the messages and information received (i.e. instructing the audience to avoid substandard and falsified medical products without ensuring that they have access to quality-assured medical products). This also has the potential to widen health inequities.

Conducting formative research and engaging communities early in the planning process allows for the identification of real barriers – whether they relate to knowledge, attitudes, infrastructure, trust or systemic inequities – and ensures that campaign strategies are appropriately aligned with the context.

People-centred, community-led and equity-focused approaches are fundamental to designing and delivering effective RCCE campaigns

Mass media campaigns tend to fail if not accompanied by or be part of a proper RCCE strategy or plan, since they reach only mainstream society and not the populations at risk.

Without the involvement of the community, communication efforts will not reach their intended audience. When done well, RCCE should help the public to enhance their protection, raise awareness about various hazards and diseases, improve health literacy, close information gaps, and combat mis- or disinformation in a timely and adaptable manner. Achieving these outcomes depends on the participation of the population and active and inclusive engagement of diverse communities.

Marginalized groups are especially important to consider as they typically face more barriers to understanding and acting on public health advice or behaviour change. They may be less likely to encounter or be able to understand campaign materials (e.g. due to language or literacy barriers) and are less likely to have the capacity to act meaningfully on public health advice due to social or economic factors. Ensuring equity is essential and RCCE campaigns should be further tailored to the unique needs and circumstances of these marginalized groups to avoid exacerbating health inequalities.

RCCE should be informed by data and evidence

Social-behavioural data and community insights are essential for developing RCCE strategies and campaigns as they inform the development of evidence-based RCCE strategies and plans. The data and insights collected promote better decision-making and can allow for stronger risk assessments by bringing a community lens to the understanding of risk during an emergency or incident involving substandard and falsified medical products. The collection, analysis and use of social-behavioural data and community insights in the design and implementation of an RCCE strategy support better response – one aligned with and appropriate to community expectations, needs and priorities.

All materials used should be culturally appropriate to the audience, in the language they understand and received through the channels they use

RCCE outreach should be culturally and linguistically adapted, as well as leverage the media types widely utilized by their intended audience, especially those who are most at risk within the population. There are many media options available, and each comes with trade-offs in cost, reach and impact that require careful consideration. At the core of these decisions should be the question: which channels are most likely to reach the highest risk groups in a format and language they can understand? It must be appropriate for local audiences, based on a strong understanding of how the audience communicates, various social and cultural characteristics which define how the audience receives information such as language spoken, levels of literacy and education and more, and access to and use of different platforms or channels.

It is essential to identify a singular focus and have a robust MEL framework

RCCE campaigns need to target a specific behaviour with clearly defined actions. Equally important is the inclusion of a strong MEL component to support continued learning for further insights and new knowledge that can be applied in future planning and strategy. Attempting to promote multiple behaviours simultaneously may dilute the message or overwhelm the audience, reducing overall effectiveness of the RCCE campaign. This singular, measurable behavioural goal should be based on evidence and community insights. This enables implementers to track progress, assess impact and identify challenges or unintended consequences in real time. A robust MEL system facilitates course correction and ensures accountability, ultimately improving the effectiveness and sustainability of the campaign.

Way forward and potential application

Future efforts to reduce public exposure to substandard and falsified medical products should build on these lessons learned, particularly when the intention is to run a risk communication campaign. Prior to a decision about whether to launch such a campaign, the following questions should be addressed:

- To what extent is lack of awareness/education the problem? Substandard and falsified medical products risk communication campaigns can only be the solution to the extent that there is a behavioural issue caused by lack of awareness or knowledge among the affected populations.
- Who is the target audience, and what media channel would be most effective in reaching and being understood by the target audience?
- Are the campaign's messages ones that the target audience can realistically act on? What parallel investments might be required to increase the capacity to act (e.g. access to health professionals and quality-assured medical products; supply side measures to improve supply chain integrity)?
- Which populations groups are most and least likely to be positively impacted by the campaign? If already vulnerable groups are the least likely to receive and benefit from the campaign, the result might be a widening of health inequities.

Any future campaigns should be robustly and independently evaluated, in order to establish the following:

- Does the campaign represent good value for money (in terms of impacts)?
- Is the campaign equitable? Are disadvantaged groups more or less likely to benefit?
- Is the campaign sustainable? What resources and other inputs are required?
- What parallel investments are required to maximize the campaign's effectiveness?

Case study 2: The role of structured risk assessment in safeguarding public health: lessons from Pakistan's 2012 drug crisis

In early 2012, over 230 cardiac patients in Lahore, Pakistan, died after consuming contaminated isosorbide mononitrate tablets distributed by the Punjab Institute of Cardiology. The tablets were found to contain pyrimethamine, a cytotoxic antimalarial drug, which caused fatal bone marrow suppression. This tragedy exposed critical gaps in Pakistan's pharmaceutical regulatory system, including the absence of early warning mechanisms, inadequate pharmacovigilance, and poor coordination between federal and provincial authorities (67–69).

Despite the rapid onset of symptoms – bleeding disorders, severe anaemia and death – the severity of the incident was not immediately recognized. The lack of a structured risk assessment framework meant that the event was not prioritized appropriately, delaying life-saving interventions. A formal risk scoring matrix could have transformed the response. If applied early, the incident would have been classified as “critical” based on:

- Likelihood of exposure: very high (the contaminated drug was administered to thousands of cardiac patients in a single hospital).
- Severity of harm: very high (patients exhibited severe haematological symptoms and high mortality within days).

This scoring would have triggered immediate public health actions, including:

- RCCE through public alerts and advisories to prevent further use of the drug.
- Stakeholder coordination by mobilizing health authorities, hospitals and pharmaceutical regulators.
- Provision of safe alternatives by rapid distribution of uncontaminated cardiac medications to affected patients.

The incident led to the creation of the Drug Regulatory Authority of Pakistan in late 2012, which introduced a national pharmacovigilance system, regional drug testing laboratories, enhanced GMP inspections and regulatory oversight. Although the reforms were reactive, not proactive, they highlight the need for embedded risk assessment protocols in pharmaceutical governance.

Lessons learned

- Structured risk assessment is essential for timely detection and prioritization of public health threats.
- A rapid scoring matrix can guide decision-makers in deploying RCCE and mitigation strategies.
- Proactive systems save lives – reactive reforms, while necessary, often come too late.

Case study 3: Global suspension of ranitidine medicines: precautionary risk assessment and RCCE in the ranitidine NDMA contamination case

In 2019–2020, regulatory agencies from the Americas and Europe detected N-nitrosodimethylamine (NDMA) impurities in ranitidine, a widely used antacid. NDMA is a probable human carcinogen, and its presence raised concerns about long-term cancer risk among chronic users. Unlike acute contamination events, the ranitidine case involved uncertainty about the contamination mechanism, variability in NDMA levels, and probabilistic harm over time (70).

Despite the lack of definitive toxicological thresholds or a clear contamination pathway, regulators applied a precautionary principle, guided by structured risk assessment. They looked at the *likelihood of exposure*, which was high as ranitidine was used by millions of patients globally, often for chronic conditions, against the *severity of harm* which was moderate to high in this scenario as the harm was long-term and probabilistic (cancer risk), but serious enough to warrant action.

This risk matrix justified a global suspension of ranitidine products, prioritizing public safety over commercial or logistical concerns. Regulators embedded into the risk management strategy to mitigate secondary harms such as untreated acid reflux or anxiety-driven medication avoidance, as well as

- Balance urgency with reassurance – messages emphasized the precautionary nature of the suspension, avoiding panic.
- Prevent abrupt discontinuation – patients were advised not to stop medication without consulting health care providers.
- Promote safe alternatives – clear guidance was provided on switching to other antacids like famotidine or omeprazole.

Lessons learned

- Awareness that this two-pronged approach maintains public trust, ensures safe transitions, and prevents both direct health risks and indirect harms from misinformation or fear, reinforcing that RCCE must be integrated into risk frameworks to ensure proportional, context-sensitive communication.
- Precautionary risk assessment is vital when scientific certainty is lacking but potential harm is significant and that effective substandard and falsified medicine risk management requires assessing both chemical and toxicological risks, and public perception and behavioural responses.

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Annex: checklist for medicines purchased online

Checklist for medicines purchased online

- ☐ Is it exactly the medicine ordered?
- ☐ Is it the correct dosage?
- ☐ Is the packaging in good condition, clean, with a patient information leaflet and in the language in which it was advertised?
- ☐ Does the medicine look, feel and smell as it should?
- ☐ Are security seals intact with no signs of tampering?
- ☐ Do any customs declarations or postal labels declare the contents as medicines?
- ☐ Do the batch number and expiry date on the primary internal packaging match the batch number and expiry date on the secondary (external) packaging?
- ☐ Have you noticed any unusual activity on your credit card since the purchase?
- ☐ Does the package or insert contain a barcode page?

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